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Exploring Nonmotorized Travel in New York State Using 2017 National Household Travel Survey Data



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November 2025

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November 2025

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ABBREVIATIONS AND ACRONYMS

4Ws	Who, Where, When, and Why
ALICE	Asset Limited, Income Constrained, Employed
FHWA	Federal Highway Administration
MOE	Margin of Error
NHTS	National Household Travel Survey
NYC	New York City
NYS	New York State
NMT	Nonmotorized
MT	Motorized
ORNL	Oak Ridge National Laboratory

EXECUTIVE SUMMARY

This report presents systematic analysis of nonmotorized travel behavior within New York State (NYS) utilizing 2017 National Household Travel Survey data. As walking and biking assume increasingly prominent roles in advancing active transportation objectives, accessible mobility, and sustainable transportation systems, comprehensive understanding of these modal patterns becomes essential for evidence-based policy development and strategic planning initiatives. Recognizing the substantial geographic, demographic, and socioeconomic heterogeneity characterizing NYS—spanning from the concentrated urban fabric of New York City (NYC) to dispersed suburban and rural contexts—this analysis employs the structured "4Ws" analytical framework to examine participant demographics, spatial and temporal distributions, and motivational factors underlying active transportation choices relative to motorized alternatives. This methodological approach captures modal behavior patterns, user characteristics, trip purposes, and temporal variations across the state's diverse contexts. Additionally, the research examines behavioral differences among distinct user classifications, including walk-only and bike-only travelers, while conducting comparative analysis of nonmotorized travel patterns between ALICE (Asset Limited, Income Constrained, Employed) and non-ALICE household categories.

The “4Ws” of Nonmotorized Travel

- Nonmotorized trips are found to constitute 25% of all travel within NYS, substantially exceeding the national average of 17%. NYC is observed to generate 71% of all nonmotorized trips statewide, reflecting the city's pronounced walkability orientation and extensive active travel infrastructure.
- Middle-aged adults (40–64 years) are found to represent 36% of nonmotorized users statewide and comprise 45% of all biking participants. Conversely, seniors (65+) and children under 16 are observed to have lower overall nonmotorized trip rates.
- Self-reported health status is found to exhibit positive correlation with nonmotorized trip frequency. Individuals engaging in light-to-moderate and vigorous physical activity are observed to have higher nonmotorized travel rates compared to those with sedentary lifestyles.
- While automobile and transit modes are found to exhibit characteristic bimodal peaks during morning and evening commute periods, nonmotorized travel is observed to show gradual morning increases with peak activity around midday before declining through evening hours.
- Weekend nonmotorized travel patterns are observed to begin later in the morning, maintain elevated levels until 3:00 p.m., then decline while remaining consistent between 3:00 p.m. and 9:00 p.m. This temporal distribution suggests increased discretionary travel for recreational, commercial, and social purposes.
- Recreational purposes are found to constitute the predominant motivation for nonmotorized trips across all geographic regions, accounting for 29% of weekday trips and 39% of weekend trips.

Comparison of Nonmotorized Travel by User Group

- Among walk-only travelers, women constitute a slight majority, representing 54% of trips. Conversely, the bike-only category demonstrates pronounced male dominance, with males accounting for 68% of bike-only trips compared to 32% for females.

- Households with children represent the largest demographic segment across all user categories (42%–50%).
- Regarding trip frequency, the bike-only group exhibits a distinct peak among middle-aged adults, with rates increasing from 2.5 among youth (5–15 years) to 4.1 and 4.3 for adults aged 25–39 and 40–64 years, respectively, before declining to 3.0 for seniors.
- Zero vehicle households are found to have significantly elevated walking trip frequencies in both NYC and the rest of NYS.
- Individuals with shorter travel times (under 30 minutes) are found to have higher biking frequencies.

Comparison between ALICE and non-ALICE Households

- ALICE households are found to exhibit consistently higher reliance on nonmotorized travel compared to non-ALICE households. In NYC, ALICE households are observed to make 48% nonmotorized trips of all trips compared to 40% for non-ALICE households. This difference becomes more pronounced in remaining NYS regions, where nonmotorized trips are found to account for 20% of total trips for ALICE households versus only 9% for non-ALICE households.
- Zero vehicle households are found to generate the highest proportion of nonmotorized trips statewide regardless of ALICE status. While this pattern holds for NYC, ALICE households are observed to have substantially higher nonmotorized trip shares (56%) compared to non-ALICE households (36%) in remaining NYS regions.
- Trip purposes are found to vary between ALICE and non-ALICE households in NYS. For school/church purposes, ALICE households are observed to have higher concentrations of nonmotorized trips during morning hours, whereas non-ALICE households are found to show a higher midday spike (12:00–2:00 pm).

1. INTRODUCTION

1.1 BACKGROUND

Walking and biking, collectively known as nonmotorized or active transportation, are increasingly recognized as essential components of a balanced and inclusive transportation system. These modes offer economically accessible alternatives to automobile travel, promote public health through physical activity, and contribute to reduced congestion on roadways. Data from the 2017 National Household Travel Survey (NHTS) indicate that walk and bike trips represent approximately 17% of all trips nationally, equivalent to about 42 billion trips annually (FHWA, 2020). However, national aggregates obscure substantial spatial variation in active transportation utilization. Certain states and metropolitan regions demonstrate significantly elevated rates of nonmotorized travel, likely reflecting the influence of built environment characteristics, transit connectivity, population density, cultural factors, and active transportation infrastructure quality (Piatkowski et al., 2019).

Examining nonmotorized transportation patterns in New York State (NYS) demands particular attention due to the state's considerable geographic, socioeconomic, and demographic heterogeneity. The state contains diverse built environments ranging from the concentrated urban fabric of New York City (NYC) to dispersed rural communities throughout upstate regions, each generating distinct travel behaviors and modal preferences. In metropolitan areas such as NYC, walk and bike infrastructure supports these modes as principal transportation options, whereas in suburban and rural contexts, walking and biking frequently function as access modes for longer-distance travel or serve recreational purposes. Beyond mobility provision, nonmotorized travel contributes to public health outcomes and transportation network adaptability. Comprehensive analysis of these phenomena requires systematic investigation extending beyond simple participation metrics to examine the demographic characteristics of users, spatial distribution of activities, temporal patterns of trip-making, and underlying motivational factors driving mode selection.

The COVID-19 pandemic underscored the critical importance of active transportation modes within urban mobility systems. During periods of reduced motorized travel and transit service disruptions, walking and biking demonstrated remarkable resilience as viable transportation alternatives, particularly in dense urban environments. Zhang and Fricker (2021) observed that nonmotorized travel not only maintained pre-pandemic levels but exhibited growth in areas with established walking and biking infrastructure. This finding aligns with broader empirical evidence demonstrating that infrastructure provision and design quality, including protected cycling facilities, continuous pedestrian networks, and traffic calming measures, serve as primary determinants of active transportation adoption and sustained usage patterns (Dill & McNeil, 2013).

Active transportation patterns exhibit pronounced associations with socioeconomic status and mobility access. Economically burdened households, including those categorized as ALICE (Asset Limited, Income Constrained, Employed) (United Way, 2025), could have higher dependence on walking and biking for essential travel due to constrained mobility options rather than discretionary choice. These populations frequently confront structural mobility barriers including limited vehicle ownership, elevated transportation cost burdens relative to household income, and spatially dispersed travel patterns requiring complex trip-chaining behaviors. Consequently, examining active transportation utilization among ALICE households provides critical insights into the broader spectrum of nonmotorized travel motivations and constraints throughout NYS.

This analysis utilizes data from the 2017 NHTS, a nationally representative cross-sectional dataset capturing daily travel behavior across US households. NYS's participation as an add-on partner involved supplemental sampling investments to achieve statistically robust representation at the state level. This enhanced sampling design facilitates granular examination of travel patterns across geographic regions, sociodemographic strata, and trip types within the state. The survey instrument captures trip-level attributes including modal choice, trip purpose, temporal characteristics, and trip duration, complemented by individual and household socioeconomic and demographic variables, thereby providing comprehensive coverage of daily mobility patterns.

To examine the behavioral dynamics of nonmotorized travel within NYS, the analysis employs the systematic "4Ws" framework to address fundamental questions regarding nonmotorized travel patterns: *Who* engages in active transportation modes, *Where* these trips occur spatially, *When* they are undertaken temporally, and *Why* individuals select walking or biking over motorized alternatives. The study incorporates modal segmentation analysis, categorizing the population into distinct user groups, such as walk-only users, bike-only users, and those who use a mix of motorized and nonmotorized modes, thereby revealing behavioral heterogeneity across travel patterns. Additionally, the research conducts comparative analysis of nonmotorized travel behavior between ALICE and non-ALICE household classifications.

This study is a continuation of a series of special population analyses that use the 2017 NHTS data to investigate distinct demographic and mobility segments. Previous installments examined rural residents, older adults, low-income communities, millennial populations, and zero vehicle households (Liu et al., 2022; Pan et al., 2024; Patwary et al., 2025a,b; Uddin et al., 2024). A comprehensive study is also conducted to analyze travel patterns, vehicle ownership trends, and overall travel behaviors in NYS regions and compare with the rest of US (Pan et al., 2025). Building upon this established analytical framework, the present study advances understanding of active transportation patterns within NYS while providing analytical inputs for enhanced transportation planning and policy development.

1.2 OBJECTIVES

This study aims to advance the understanding of nonmotorized travel behavior in NYS across different population groups. The analysis provides comprehensive characterization of nonmotorized mobility utilizing detailed trip-level data from the 2017 NHTS. The specific research objectives include:

- Examine the “4Ws” of nonmotorized travel: identify who makes nonmotorized trips, where these trips occur across the state, when they happen during the day and week, and why they are undertaken
- Analyze characteristics of nonmotorized travelers by gender, age, income, and household composition
- Explore key patterns and characteristics of nonmotorized travel, in terms of daily trip rate
- Compare nonmotorized travel behavior between ALICE and non-ALICE households.

1.3 DATA

1.3.1 NHTS

The NHTS is a comprehensive dataset that captures detailed information on travel behavior across the US. It includes data on trip purpose, mode, distance, duration, and socio-demographic characteristics of both individuals and households. NYS participated in the 2017 NHTS as a state add-on, resulting in a substantial sample size that allows for robust disaggregated analyses. This enables a focused examination of how daily travel behaviors differ across population subgroups within NYS (NHTS, 2017).

In 2017, the survey included more than 129,000 households: 26,000 from the national sample plus additional respondents from 13 state or metropolitan planning organization add-ons. As part of the NYS add-on, more than 17,000 households across the state were surveyed, making it one of the largest subsamples within the 2017 dataset. This enables us for a more granular look at nonmotorized travel behavior in the region. Notably, the 2017 survey was the first NHTS cycle to fully adopt an online survey platform, with enhanced GIS capabilities for household and trip-level attributes. This provides a rich foundation for examining behavioral trends, multimodal patterns, and emerging mobility dynamics.

1.3.2 ALICE

The ALICE household classification has garnered increasing scholarly and policy attention as a framework for examining economic vulnerability and transportation accessibility challenges. This analysis incorporates data from the United Way's ALICE research initiative (United Way, 2025) to examine socioeconomic influences on travel behavior. The ALICE classification identifies households earning above federal poverty thresholds yet having trouble affording basic living expenses including housing, transportation, childcare, healthcare, and food costs. Within NYS, approximately half of all households meet ALICE criteria. These populations frequently experience distinctive transportation constraints stemming from limited financial resources, restricted vehicle ownership, and constrained modal alternatives, rendering their mobility patterns critical for comprehensive transportation planning and policy development.

1.4 USER GROUPS

To better understand the diversity of daily travel patterns and participation in nonmotorized transportation, individuals were classified into five user groups based on transportation modes they used on their assigned travel day. This modal segmentation methodology facilitates detailed analysis of multimodal travel behavior while enabling comparative assessment across demographic and socioeconomic variables.

- **Walk-only.** Individuals who completed at least one walking trip but did not report any bike trips on their travel day. These individuals may have also used motorized modes (e.g., driving or transit) in combination with walking.
- **Bike-only.** Individuals who made at least one bike trip but no walking trips. As with the walk-only group, bike-only users may have also used motorized transportation on the travel day.
- **Nonmotorized.** A broader category that includes individuals who made at least one walk or bike trip, regardless of whether they also used motorized modes. This group captures all forms of active travel participation.

- **Motorized.** Individuals who completed one or more trips using motorized modes (e.g., private vehicle, public transit, taxi/rideshare). This group includes those who may have also walked or biked on the travel day.
- **All.** The complete set of survey respondents who reported making any trip on the travel day, regardless of mode.

1.5 REPORT ORGANIZATION

This report presents a systematic assessment of nonmotorized travel behavior in NYS, examining modal patterns, user demographics, spatiotemporal distributions, and socioeconomic disparities in travel behavior through analysis of 2017 NHTS data. Section 2 investigates the fundamental dimensions of nonmotorized travel: spatial distribution of trip origins and destinations, temporal patterns across daily and weekly cycles, demographic characteristics of participants, and underlying trip motivations. Section 3 examines user segmentation and behavioral heterogeneity through comparative analysis of travel characteristics across five modal classifications (All users, Motorized, Nonmotorized, Walk-only, and Bike-only); thereby revealing variation in daily mobility patterns and multimodal integration throughout the state. Section 4 conducts detailed comparative analysis of nonmotorized travel behavior between ALICE and non-ALICE household categories, examining socioeconomic influences on active transportation choices. The report concludes with synthesis of primary findings, discussion on implications for transportation planning practice, and identification of future research priorities.

2. THE 4WS OF NONMOTORIZED TRAVEL

The analysis in this Section employs a systematic "4Ws" framework to investigate four fundamental dimensions of nonmotorized travel: *Who* engages in nonmotorized trips, *Where* these trips occur, *When* they are undertaken, and *Why* they are made. This comprehensive framework enables a multidimensional examination of nonmotorized travel behavior across demographic, spatial, temporal, and motivational domains, providing insights into the complex factors that influence active transportation choices.

2.1 Who?

2.1.1 Age of Persons Reporting Nonmotorized Travel

Who participates in nonmotorized travel is a critical first step to understanding the dynamics in active travel. It has been observed from 2017 NHTS data that one in every four trips (25%) is nonmotorized (either by walking or biking) in NYS. This rate is significantly higher than the national average of 17% (FHWA, 2020). Figure 2-1 shows the age distribution of persons reporting nonmotorized trips, either walk, bike or both modes. Among NYS residents reporting at least one nonmotorized trip, middle-aged adults (40–64 years) constitute the largest demographic segment, representing approximately 36% of all nonmotorized trip-makers. This predominance likely reflects multiple contributing factors: sustained physical mobility and health status within this age cohort, discretionary travel choices for utilitarian and recreational purposes, and increased residential proximity to walkable built environments. The prominence of this demographic becomes more pronounced when examining biking behavior specifically, with adults aged 40-64 comprising nearly 45% of all bike trip-makers statewide.

Conversely, seniors aged 65 and older and children under 16 make up the smallest share of nonmotorized trip-makers. For seniors, this could be attributed to physical limitations, safety concerns, or lack of accessible infrastructure. For children, trip dependency and adult supervision needs may limit their ability to engage in independent walking or biking. Interestingly, younger adults aged 16 to 24 show lower participation than expected. While they do contribute meaningfully to nonmotorized travel, their share is still outpaced by the 25–39 and 40–64 age groups.

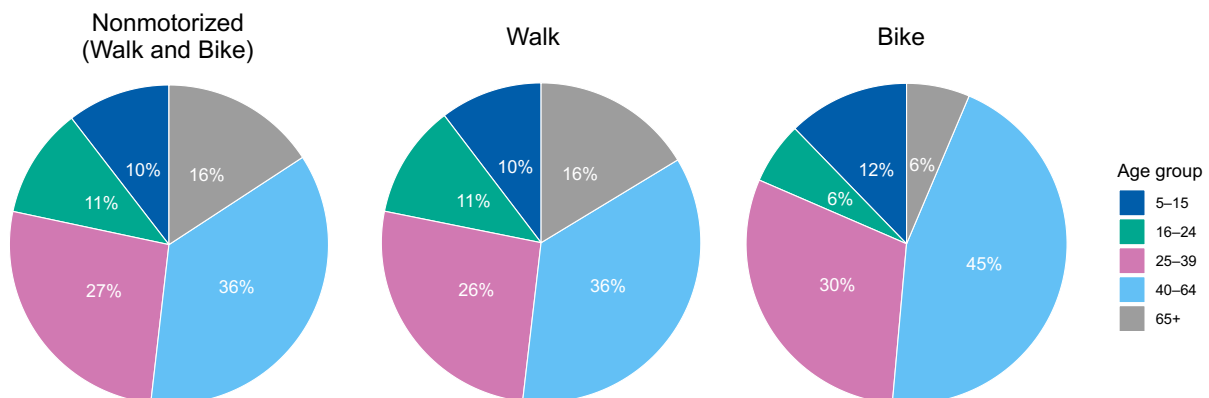


Figure 2-1. Age Distribution of Persons Reporting Nonmotorized Trips in NYS

2.1.2 Health Status

Figure 2-2 illustrates the share of nonmotorized trips across different self-reported health statuses. The relationship is clear: better health is strongly associated with higher levels of nonmotorized travel. Individuals who rated their health as “Excellent” account for the largest share of nonmotorized trips (38%), followed by those reporting “Very good” (31%) and “Good” (22%) health. In contrast, the share of nonmotorized trips drops significantly among those with “Fair” health (7%) and is nearly negligible among those who rate their health as “Poor” (1%). This distribution highlights the role of physical ability in nonmotorized travel choices. Healthier individuals may be more capable and confident in walking or biking, while those with poorer health may face limitations that restrict such travel. It also underscores a potential loop: active modes like walking and biking may help individuals maintain or improve their physical well-being, while those in poor health may be excluded from the benefits of active transportation due to existing health barriers.

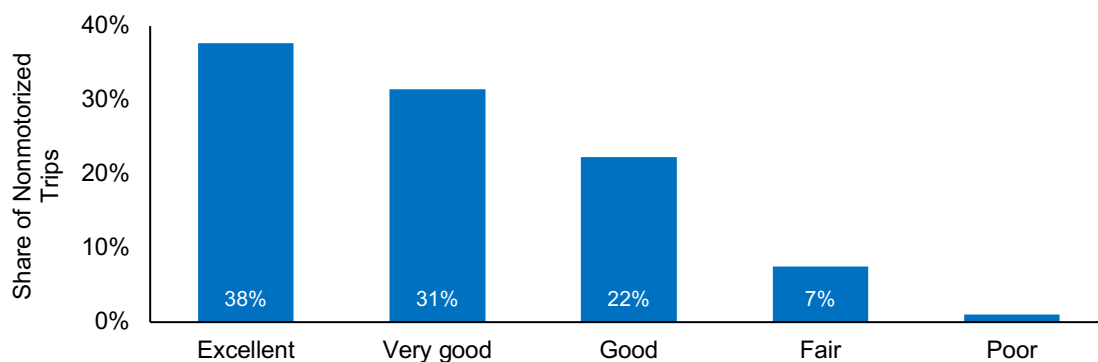


Figure 2-2. Nonmotorized Travel by Health Status in NYS

2.1.3 Physical Activity

Figure 2-3 demonstrates a clear association between lifestyle physical activity levels and nonmotorized travel participation. Individuals reporting light to moderate physical activity account for 60% of all nonmotorized trips, while those engaging in vigorous physical activity contribute 34% of trips. Conversely, individuals with minimal or no regular physical activity represent only 6% of nonmotorized trips. This distribution indicates a strong positive association between existing physical activity engagement and the propensity to utilize active transportation modes. The findings suggest a bidirectional relationship wherein active lifestyles predispose individuals toward nonmotorized travel choices, while regular walking and biking for transportation may reinforce and sustain active lifestyle patterns.

2.2 Where?

2.2.1 NYC vs. Rest of NYS

Figure 2-4 illustrates a contrast in nonmotorized travel behavior between residents of NYC and those living in the rest of the state. A substantial 71% of all nonmotorized trips in NYS originate from NYC, underscoring the city’s strong orientation toward walkability and active travel infrastructure. In comparison, the rest of NYS accounts for only 29% of all nonmotorized trips, despite representing a much larger geographic area. This concentration of nonmotorized travel within NYC reflects the city’s dense urban form, mixed land use, and extensive pedestrian and bicycle infrastructure. Additionally, factors such as lower vehicle ownership rates and greater access to transit in NYC likely contribute to the high

share of walk and bike trips. Conversely, the suburban and rural nature of much of the rest of the state could present challenges for active transportation due to greater travel distances and limited pedestrian and biking infrastructure.

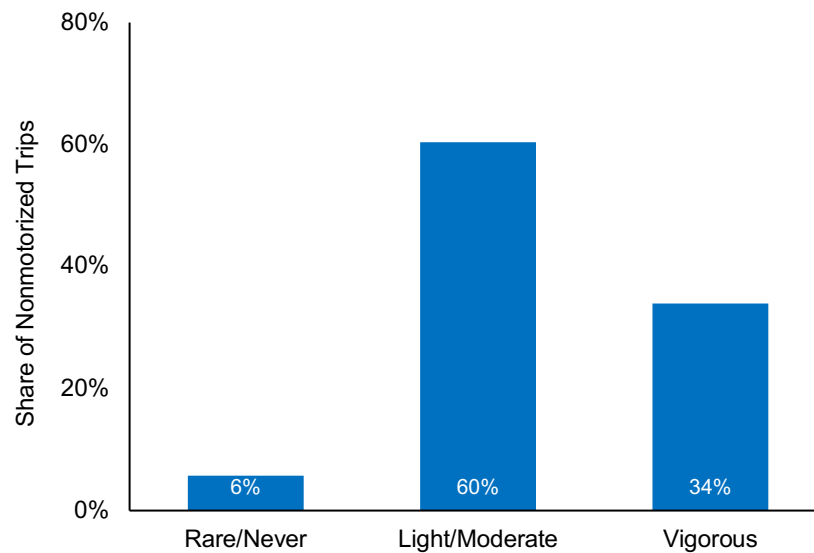


Figure 2-3. Nonmotorized Travel by Level of Physical Activity in NYS

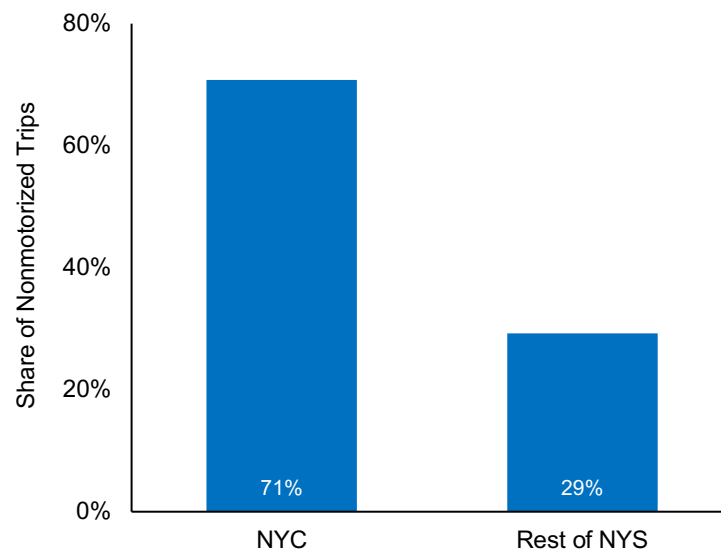


Figure 2-4. Nonmotorized Travel in NYC and the Rest of NYS

2.2.2 Home Geography

The 2017 NHTS includes a derived variable, *HBHUR*, which classifies households into five residential geography types: Urban, Suburban, Second City, Small Town, and Rural. This classification is based on

population density, land use patterns, and commuting flows at the Census Block Group level, and it provides a useful lens for analyzing travel behaviors across varying built environments. Based on this variable, Figure 2-5 shows that urban households overwhelmingly account for 74% of all nonmotorized trips in NYS, further emphasizing the strong relationship between dense urban environments and active travel modes. In contrast, suburban (9%), second city (10%), small town (4%), and rural (3%) areas contribute much smaller shares of nonmotorized travel. These results highlight the critical role that urban form and local accessibility play in supporting walking and biking; modes that are often less viable in low-density, car-dependent environments. The relatively higher share in second cities compared to other non-urban geographies could indicate some potential for improving active travel in mid-sized population centers.

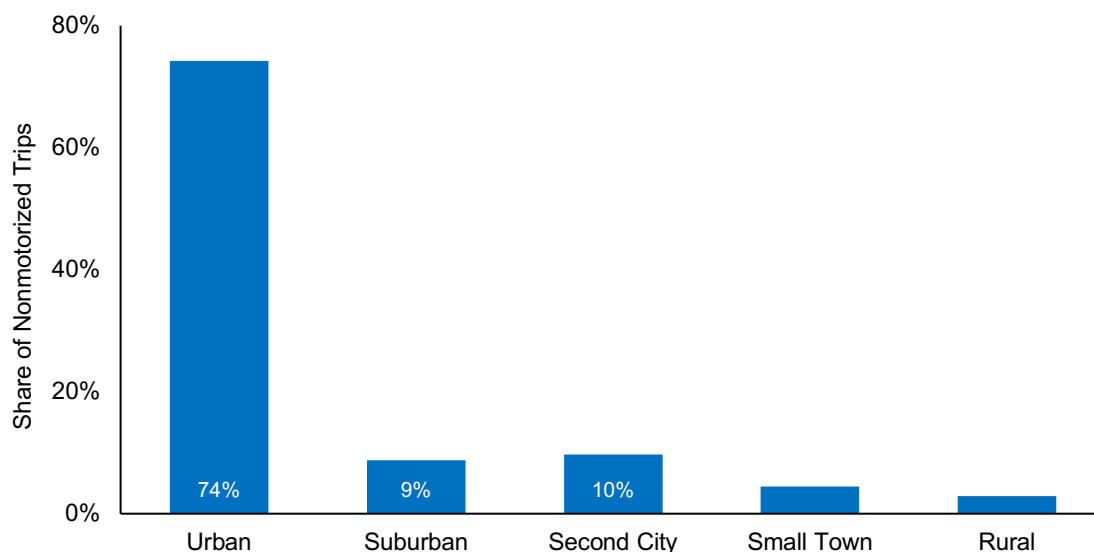


Figure 2-5. Nonmotorized Travel by Home Geography in NYS

2.3 When?

2.3.1 Time of Day

Figure 2-6 presents the distribution of trips by time of day for three travel modes in NYS: nonmotorized (walking/biking), auto, and transit. This temporal analysis sheds light on how trip-making behavior varies throughout the day depending on the travel mode. Auto travel displays two pronounced peaks: one in the morning around 8 a.m. and another in the late afternoon around 5 p.m. This pattern reflects the traditional work commute periods and indicates the dominance of auto use for employment-related travel. Transit trips show a similar dual-peak structure but with slightly earlier morning activity and a more concentrated evening peak. This suggests that transit is heavily used during commuting hours, likely due to scheduled service availability and concentration of transit-dependent populations in urban centers.

Nonmotorized travel demonstrates a distinctly different temporal distribution, characterized by relatively uniform activity levels throughout the day. Walking and biking trips exhibit a gradual morning increase, sustain consistent volumes from mid-morning through late afternoon, and decline progressively during evening hours. This pattern contrasts markedly with the pronounced fluctuations observed in motorized modes, which correspond closely to structured employment schedules. The smoother temporal profile of nonmotorized travel reflects its accommodation of diverse trip purposes—including utilitarian errands,

educational journeys, recreational activities, and access to proximate destinations—which occur with greater temporal flexibility than work-related travel. This temporal divergence underscores the critical role of active transportation in addressing mobility needs during off-peak periods and demonstrates its function as an adaptable and accessible transportation option across varied daily activity patterns.

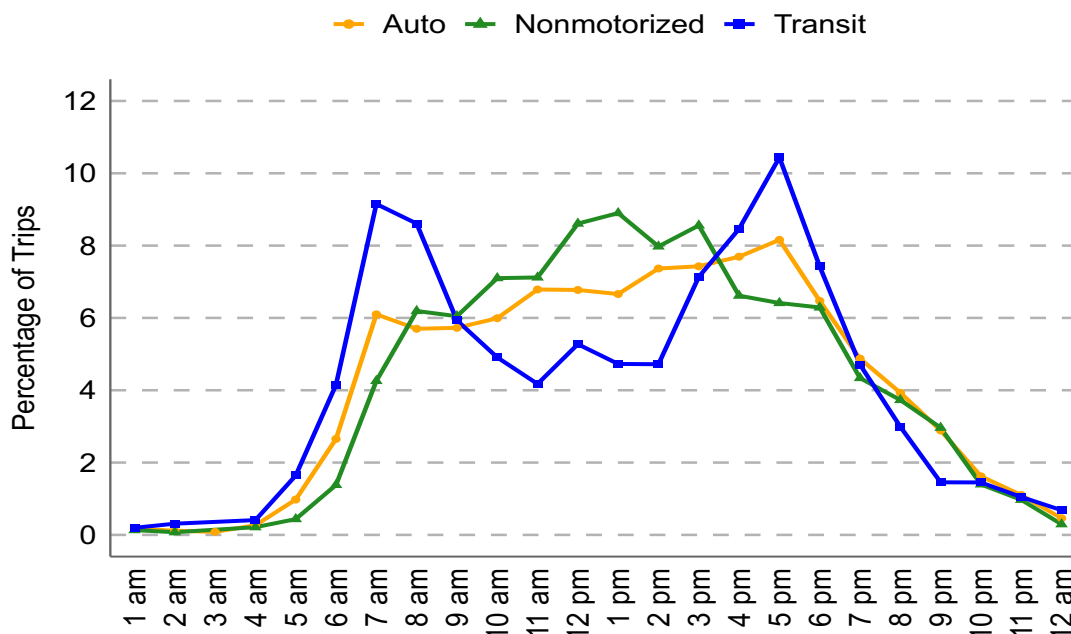


Figure 2-6. Comparison of Travel Mode by Time of Day in NYS

If look at the distribution in NYC and the rest of NYS separately, notable differences emerge in how each travel mode is used throughout the day across regions (Figure 2-7). In NYC, transit plays a dominant role during peak commute periods, with distinct spikes around 7–9 a.m. and 4–6 p.m. The afternoon peak is especially pronounced, with transit reaching its highest share near 5 p.m., highlighting the city’s dependence on public transportation. Nonmotorized travel in NYC follows a relatively steady upward trend throughout the morning, peaking around noon to 1 p.m., before gradually tapering off into the evening. This pattern indicates that walking and biking are used not just for commuting but also for a variety of midday activities such as errands, school, or leisure.

In contrast, the rest of NYS displays a higher reliance on auto travel, particularly in the afternoon, where auto and nonmotorized trips peak around 3–5 p.m. While nonmotorized travel remains active throughout the day, it is more evenly distributed without the midday concentration seen in NYC. Transit use outside of NYC also features a dual peak pattern, but its overall volume is much lower, with sharp declines after commute hours, especially in the evening, reflecting the more limited availability and use of transit options in less urbanized areas.

2.3.2 Day of Week

Figure 2-8 illustrates how nonmotorized travel varies by hour of the day, separately for weekdays and weekends in NYS. This comparison highlights clear temporal differences in trip-making behavior based on the type of day. On weekdays, nonmotorized trips follow a more structured pattern, with noticeable increases during the morning hours (7–9 a.m.), consistent with work- or school-related commuting activity. This is followed by a steady volume throughout the day and a second, more prominent peak in

the late afternoon (4–6 p.m.), likely representing after-work trips and recreational or exercise-related walking and biking. In contrast, weekend nonmotorized travel is more evenly distributed throughout the day. Trip volumes begin increasing later in the morning and remain elevated between 10 a.m. and 5 p.m., indicating a shift toward discretionary purposes such as leisure, shopping, and social activities. The midday dominance of weekend travel may suggest a more relaxed schedule and different population groups such as families or older adults engaging in walking and biking trips. This temporal differentiation and usage patterns between weekdays and weekends could have important implications for transportation planners in terms of improvements in pedestrian and bike infrastructure.

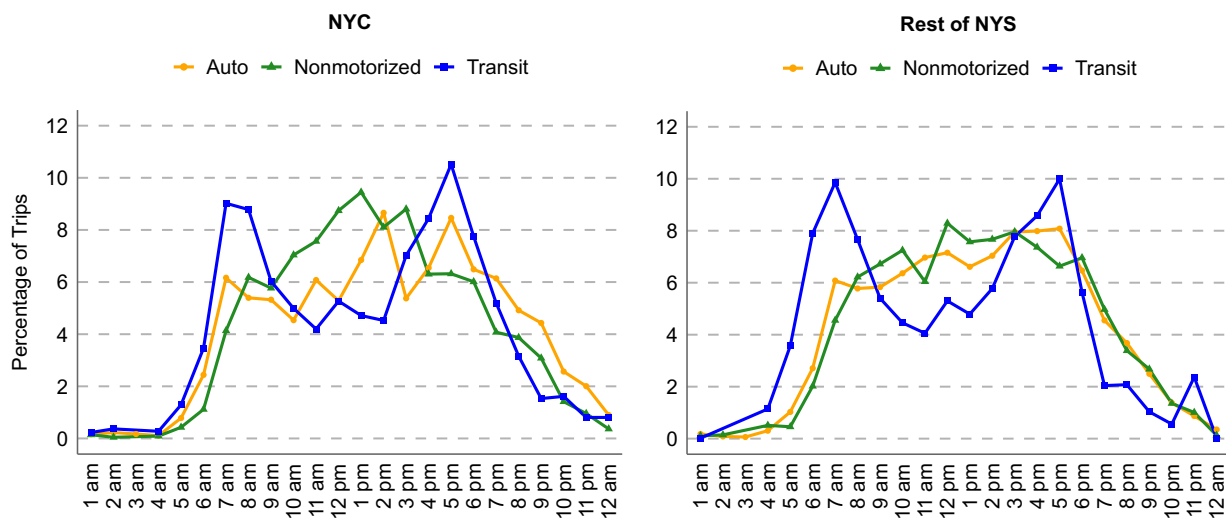


Figure 2-7. Comparison of Travel Mode by Time of Day in NYC and the Rest of NYS

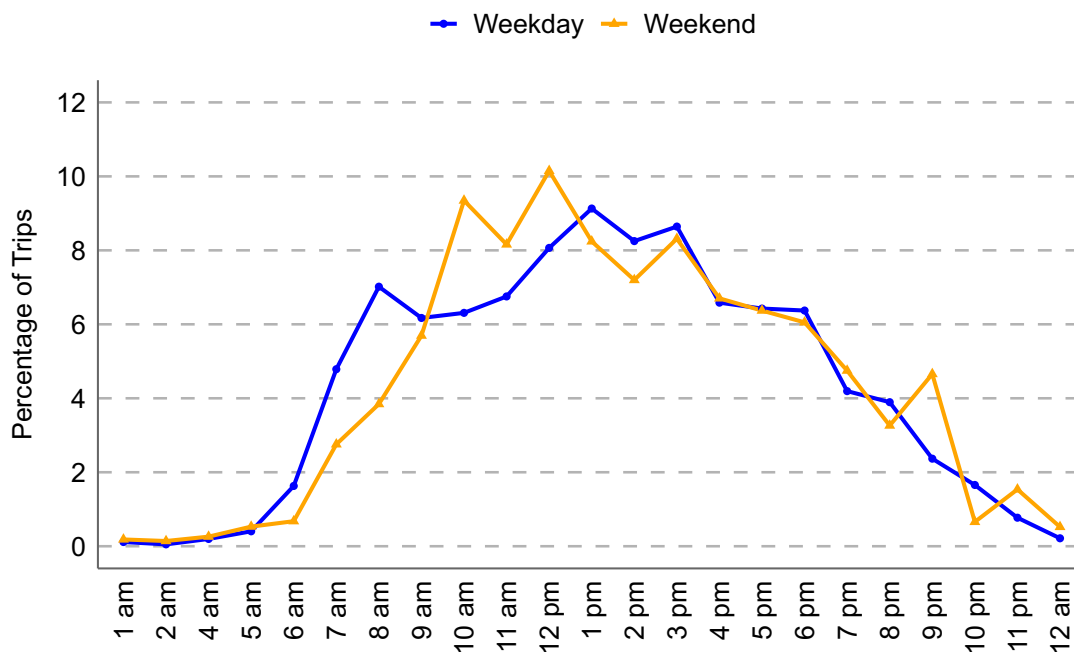


Figure 2-8. Nonmotorized Trips by Time of Day and Day of Week in NYS

When we look at the distribution of nonmotorized trips by time of day and day of week in NYC and the rest of NYS (Figure 2-9), some key differences and patterns emerge between weekdays and weekends. In NYC, nonmotorized trips exhibit a smoother distribution across the day, with relatively similar levels of activity during both weekdays and weekends. On weekdays, the trip volume starts increasing around 6 a.m., peaking slightly after 11 a.m. to noon, and then gradually tapering off into the evening. Weekend patterns are more delayed: the volume begins picking up around 8 a.m., with a more pronounced peak around midday (11 a.m. to 1 p.m.), likely reflecting recreational or discretionary trip purposes. The weekend curve shows slightly higher percentages than weekdays during the late morning to early afternoon, suggesting that walking and biking may be more commonly used for leisure or errands on weekends in NYC. In comparison, the rest of NYS displays a more distinct separation between weekday and weekend trip timing. On weekdays, there is a sharp morning peak from 6 a.m. to 9 a.m., which then drops slightly and stays relatively stable through the afternoon before tapering off. This weekday peak aligns with typical work or school commute periods. On weekends, the peak shifts later in the day; between 10 a.m. and 1 p.m., which is similar to the pattern observed in NYC. However, the weekend volumes in the rest of NYS are consistently lower than weekday volumes until the mid-morning period.

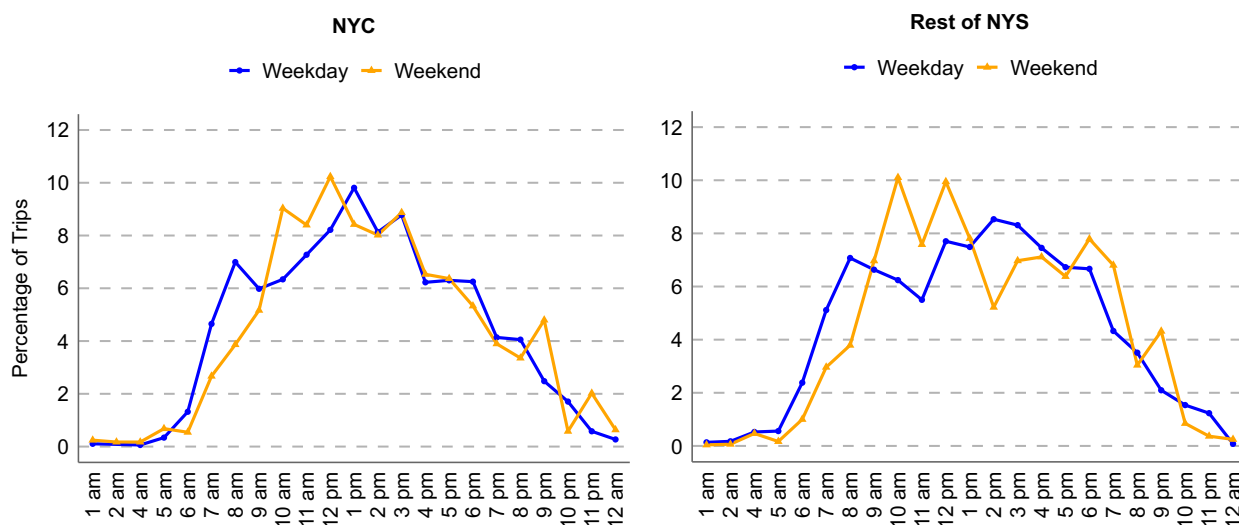


Figure 2-9. Nonmotorized Trips by Time of Day and Day of Week in NYC and the Rest of NYS

2.4 Why?

2.4.1 Trip Purpose by Day of Week

To better understand why nonmotorized trips are taken in NYS—that is, the reasons motivating walking and biking—we examined trip purposes by day of the week. Figure 2-10 shows the distribution of nonmotorized trip purposes across weekdays and weekends. On weekdays, the most common purpose for nonmotorized trips is recreational, accounting for 29% of all trips. This is followed by family and personal trips (21%) and shopping trips (18%). Work commutes make up only 11% of trips. The relatively high share of recreational and shopping trips during weekdays may reflect the role of walking and biking for shorter, local errands and leisure activities that are more flexible in timing. On weekends, the distribution shifts, with recreational trips increasing even further to 39%, becoming the dominant purpose. At the same time, the share of work trips drops sharply to 5%, consistent with lower commuting demand. Interestingly, the share of other personal businesses drops to 10% on weekends. Shopping trips increase to 24%, while school/church and others hold similar proportions between weekdays and weekends.

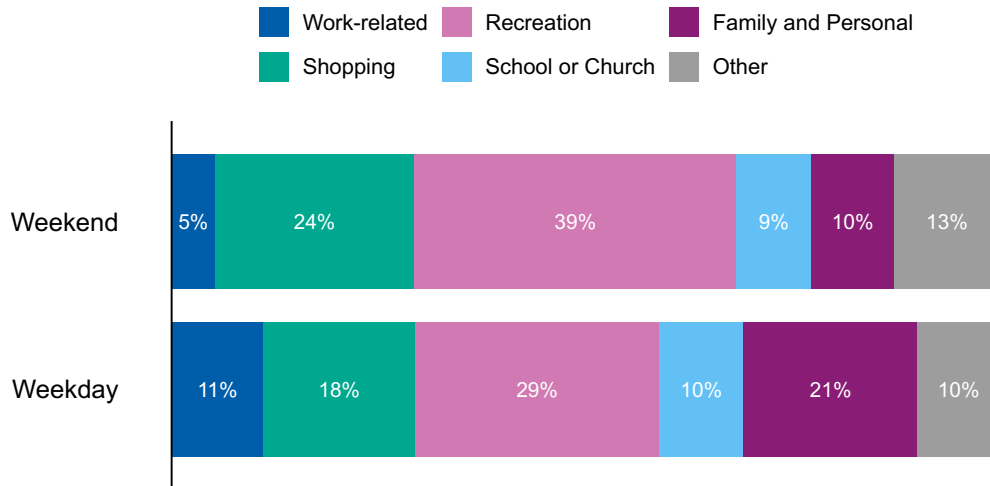


Figure 2-10. Nonmotorized Trip Purpose by Day of Week in NYS

Figure 2-11 extends the analysis by comparing nonmotorized trip purposes across weekdays and weekends separately for NYC and the rest of NYS. This regional breakdown underscores the nuanced ways in which walking and biking serve different functions across urban and non-urban contexts. In NYC, weekdays see a relatively even distribution among several trip purposes, with recreational trips making up 28%, followed by family and personal errands at 23%, and shopping trips at 20%. Work trips also hold a notable share at 12%, which is higher than the statewide average. On weekends, the share of work-related nonmotorized travel falls to just 6%, while recreational trips increase to 40%.

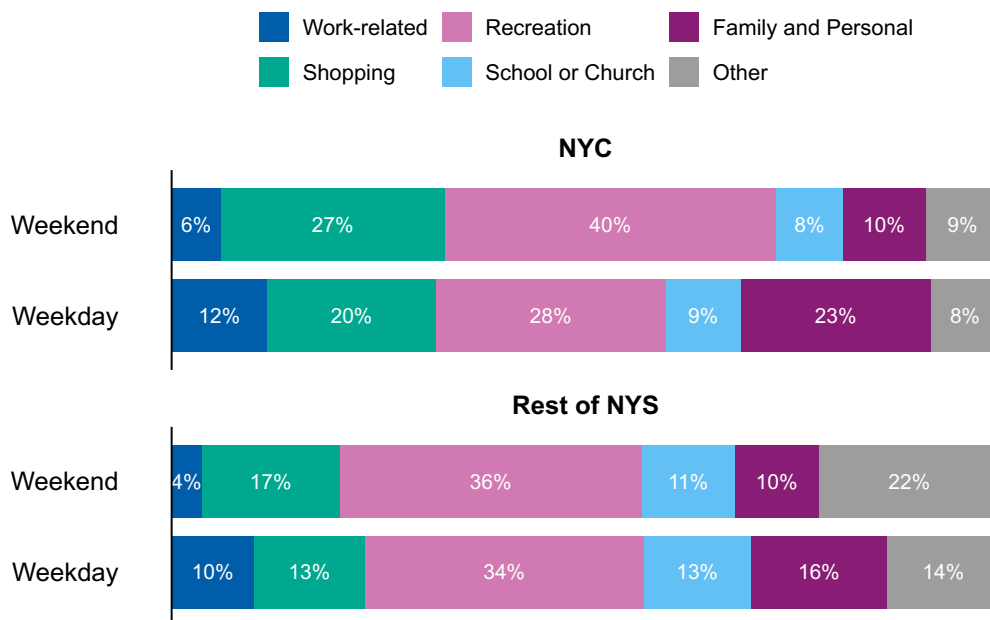


Figure 2-11. Nonmotorized Trip Purpose by Day of Week in NYC and the Rest of NYS

In contrast, the rest of NYS displays a different profile. On weekends, recreational purpose accounts for 36% of nonmotorized trips, slightly lower than NYC, while shopping and work trips make up smaller proportions at 17% and 4%, respectively. This suggests that outside the city, nonmotorized modes are used less for utilitarian purposes such as commuting or errands, and more for leisure. The “other” category also plays a more prominent role in the rest of NYS, accounting for 22% of trips, which may reflect a broader range of informal or irregular trip purposes in less dense areas. On weekdays, recreational travel remains the dominant purpose at 34%, while shopping (13%) and work (10%) follow behind. Interestingly, the share of school or church-related trips increases slightly to 13%, compared to just 9% in NYC, perhaps pointing to more structured religious or community-based weekend activities in non-urban areas.

2.4.2 Trip Purpose by Home Geography

In Figure 2-12, we explore the distribution of nonmotorized trip purposes by home geography in NYS. Across all geographies, recreational purpose accounts for the largest share of nonmotorized trips. However, the proportion varies notably by location. In rural cores, more than half (51%) of nonmotorized trips are for recreational, while this share drops to 44% in small towns and further to 42%, 31%, and 23% in suburban, urban areas, and second cities, respectively.

Work trips, on the other hand, are more common in second city and urban settings. School/church trips peak in second cities (20%) and are lowest in rural areas (6%), likely due to differences in proximity to educational or religious institutions. Shopping trips show the opposite gradient compared to recreational trips. They are lowest in rural cores (10%) and small towns (11%) but increase to 18% in second cities and peak at 22% in urban areas. Meanwhile, the share of family and personal trips is the highest in urban areas, while the “other” category is highest in rural areas.

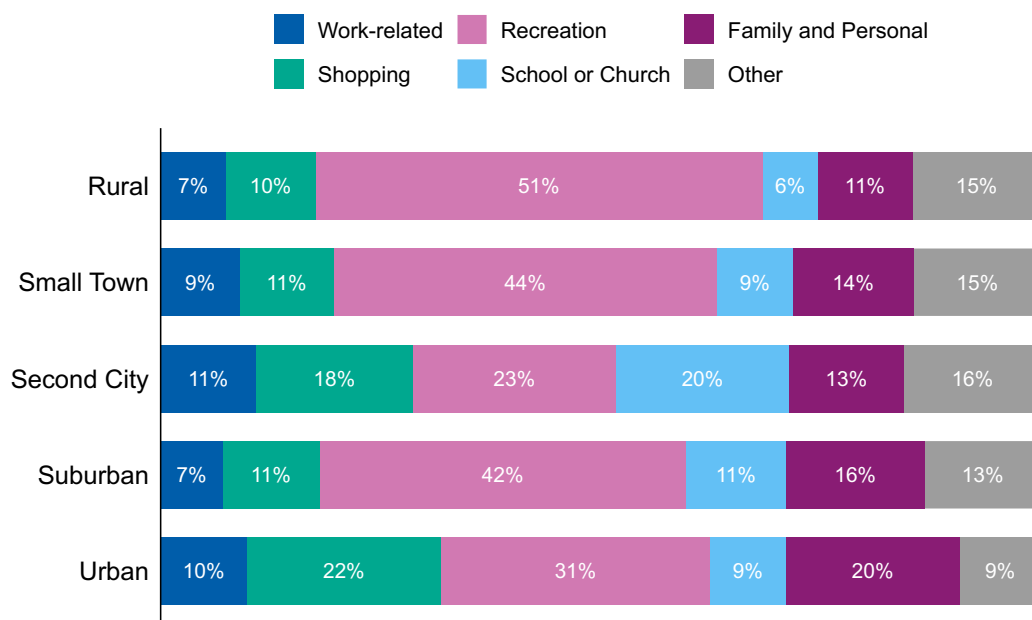


Figure 2-12. Nonmotorized Trip Purpose by Home Geography in NYS

3. COMPARISON OF NONMOTORIZED TRAVEL BY USER GROUP

To examine nonmotorized travel behavior beyond simple participation rates, this analysis employs a five-group modal segmentation framework based on daily travel patterns (Figure 3-1). The "All" group includes all respondents completing at least one trip regardless of mode, while "Motorized" users represent individuals utilizing at least one motorized mode (automobile, public transit, or other motorized transportation), potentially alongside walking or biking. The "Nonmotorized" group comprises individuals completing at least one walking or biking trip while potentially using motorized modes, capturing multimodal integration patterns. Within this category, "Walk-only" users engaged exclusively in pedestrian travel, while "Bike-only" users utilized biking without walking. This classification system enables examination of nuanced behavioral differences among active transportation participants while facilitating comprehensive analysis across demographic, socioeconomic, and geographic variables.

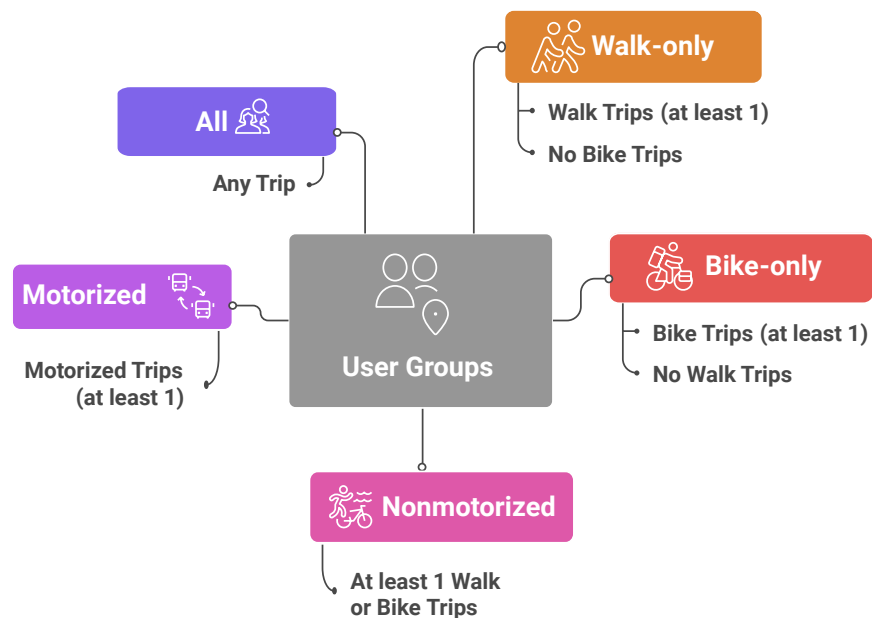


Figure 3-1. User Groups Based on Travel Mode Participation

3.1 DISTRIBUTION OF USER GROUPS

3.1.1 Sex

Figure 3-2 presents the sex distribution across the five user groups. Among walk-only travelers, women make up a slight majority, accounting for 54% of trips, while men account for 46%. In contrast, the bike-only category is strongly male-dominated. Approximately 68% of bike-only trips are made by males, while only 32% are made by females. This disparity is consistent with national findings that show women are less likely to bike (Velotricbike, 2023). When considering the broader nonmotorized group (those who either walk or bike), females still slightly outnumber males, representing 53% of all such trips. For motorized trips, the gender split is nearly even, with females accounting for 51% and males for 49%. The overall travel distribution across all modes follows a similar pattern, reflecting the general demographic balance of the population.

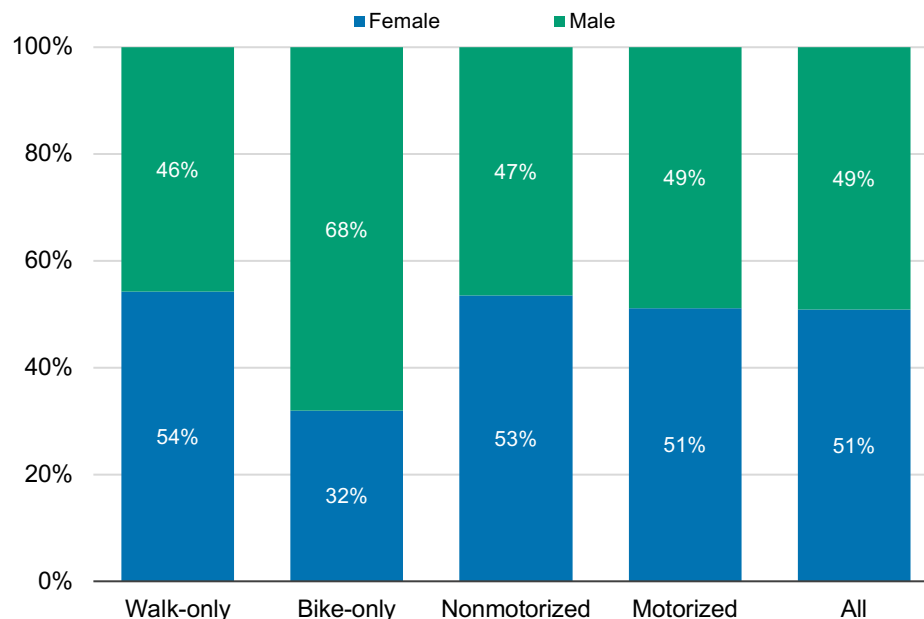


Figure 3-2. Distribution of User Groups by Sex in NYS

3.1.2 Age

Table 3-1 summarizes the distribution of the five user groups across different age categories. It can be observed that among walk-only travelers, the largest share (35%) comes from the 40–64 age group, followed closely by the 25–39 group (25%). Interestingly, the youngest age group (5–15) accounts for 13% of walk-only travelers—reflecting school-aged children who often walk to school or nearby locations. The oldest group, aged 65 and over, makes up 16% of walk-only trips. In contrast, bike-only users are skewed much more heavily toward younger and middle-aged adults. The 25–39 age group comprises the largest share of bike-only trips at 31%, followed by the 40–64 group (40%). Children aged 5–15 contribute 15% of bike-only trips, which aligns with recreational and school-related biking. Notably, the share of bike-only users drops sharply among older adults (6% for age 65+). The combined nonmotorized group (walk or bike) largely mirrors the walk-only pattern. Adults aged 40–64 make up the largest segment (35%), followed by those aged 25–39 (25%). Seniors (65+) and youth (5–15) each represent 13–16% of this group, indicating a wide age spread in nonmotorized travel.

Table 3-1. Distribution of User Groups by Age in NYS

Age	Walk-only	Bike-only	Nonmotorized	Motorized	All
5–15	13%	15%	13%	13%	13%
16–24	12%	7%	11%	12%	12%
25–39	25%	31%	25%	23%	23%
40–64	35%	40%	35%	38%	37%
65+	16%	6%	16%	15%	15%

To better understand the age-based variation in nonmotorized transport, Figure 3-3 presents the distribution of walk-only, bike-only, nonmotorized, motorized, and all users by generational cohort. This generational lens offers further insight into how travel preferences align with broader life stage

characteristics and mobility needs. Gen X (born 1965–1980) emerge as the dominant group among bike-only users, comprising 32% of that group, followed closely by Millennials (born 1981–1996) at 30%. This aligns with existing research suggesting that younger and middle-aged adults are more likely to adopt bicycling (FHWA, 2020), possibly due to their greater physical ability, openness to alternative modes, and preference for sustainable transport options. Millennials and Baby Boomers are also tied at 26% for walk-only and nonmotorized trips overall, indicating that both generations actively engage in walking as part of their travel behavior. Gen Z (those younger than Millennials) contributes 19% to walk-only and bike-only users and 20% to nonmotorized and motorized trips overall. Their mobility decisions are likely shaped by affordability, limited access to cars, and proximity to schools or jobs. Baby Boomers (born 1946–1964) also represent 26% of walk-only and nonmotorized users, mirroring their share in the motorized and overall population. This consistent share indicates that walking remains an important mode for many older adults, especially for short distances and daily needs, although biking is less common in this group (only 19%). In contrast, the oldest generation (labeled as Other Older Gen) makes up only 7% of walk-only and nonmotorized users, and just 1% of bike-only trips. Their lower representation across all categories is expected due to reduced mobility and travel frequency in advanced age.

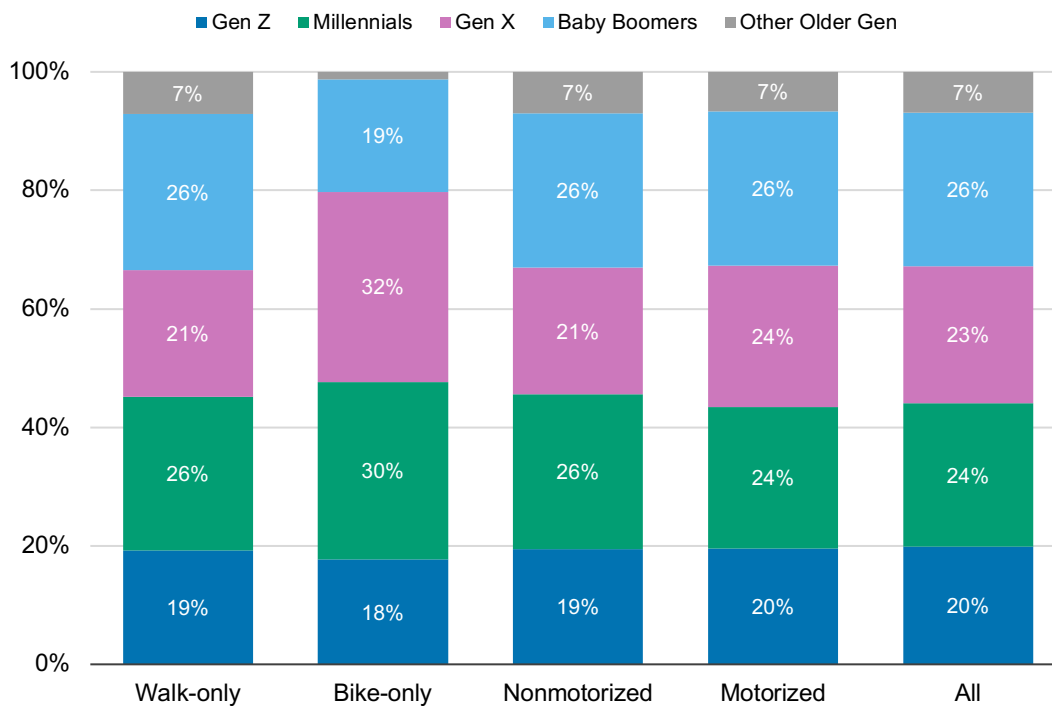


Figure 3-3. Distribution of User Groups by Generation in NYS

3.1.3 Household Size

Figure 3-4 illustrates how household size influences nonmotorized travel by showing the distribution of user groups across three household size categories: one person, two persons, and more than two persons. It indicates that individuals from larger households (with more than two persons) are more likely to fall into the motorized and all user groups, making up 60% of those groups. In contrast, among walk-only and bike-only users, just half come from households with more than two persons. Two-person households are slightly more common among bike-only users (33%) than other groups, while walk-only and nonmotorized users show identical shares (30%) in this category. Interestingly, one-person households are more represented among walk-only users (20%) compared to motorized (13%) or all users (14%).

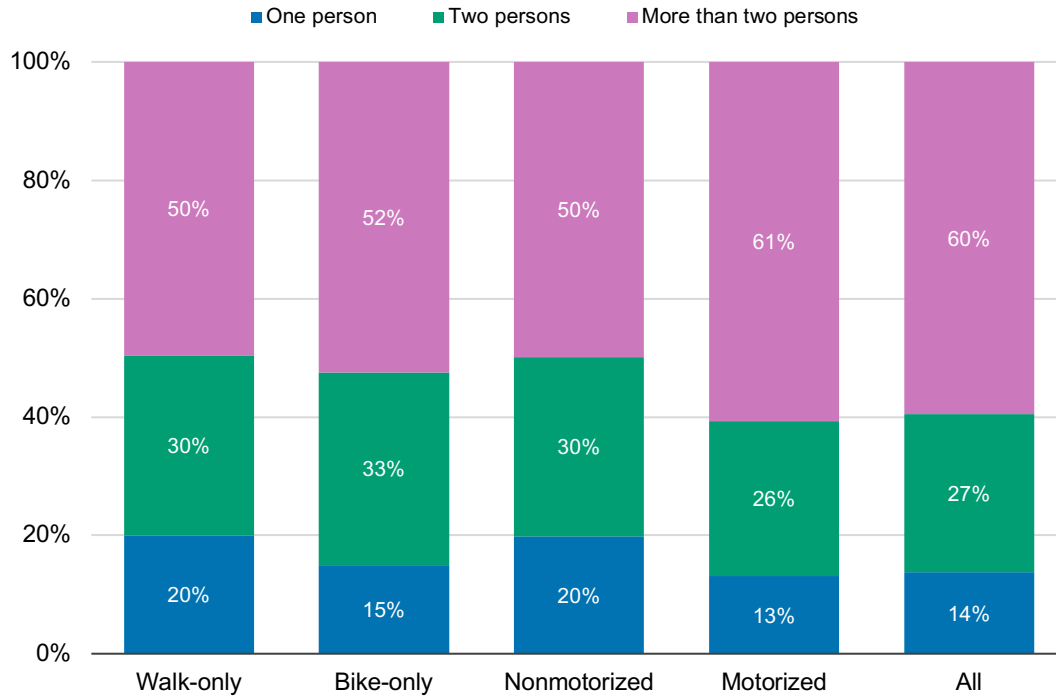


Figure 3-4. Distribution of User Groups by Household Size in NYS

3.1.4 Vehicle Ownership

Vehicle availability plays a critical role in shaping travel behavior. Table 3-2 reveals distinct differences in household vehicle ownership patterns across user groups. Walk-only and bike-only users are substantially more likely to live in households without a vehicle, 42% and 45% of these groups, respectively, report zero vehicle ownership. Similarly, 42% of the overall nonmotorized users come from zero vehicle households. In contrast, only 18% of motorized users and 22% of the all users live in zero vehicle households, underscoring the stronger reliance on private or public motorized transportation among those with vehicle access. On the other hand, households with more than two vehicles are predominantly found among motorized users (22%) and all users group (20%), compared to just 5–8% among nonmotorized user groups. This pattern suggests a clear association between higher vehicle ownership and greater dependence on motorized travel. Meanwhile, single-vehicle households are relatively consistent across all groups, comprising roughly one-third of each. These findings emphasize the mobility constraints faced by zero vehicle households and their likely dependence on walking or biking.

Table 3-2. Distribution of User Groups by Household Vehicle Ownership in NYS

Vehicle Ownership	Walk-only	Bike-only	Nonmotorized	Motorized	All
Zero vehicle	42%	45%	42%	18%	22%
One vehicle	34%	33%	33%	29%	29%
Two vehicles	16%	17%	16%	31%	28%
More than two vehicles	8%	5%	8%	22%	20%

3.1.5 Household Lifecycle

The lifecycle stage of a household has significant implications for travel behavior and mode choice. Figure 3-5 shows that households with children represent the largest share across all user groups, accounting for 42% of walk-only, bike-only, and nonmotorized users, and an even higher 50% among motorized users. This dominance is mirrored in the overall population, where nearly half of all users (49%) reside in households with children. Multiple adult households without children also represent a notable share, particularly among bike-only users (37%), followed by walk-only and nonmotorized users (both 25%). These households likely reflect working-age adults who may engage in more discretionary travel, including biking. Single adult households without children make up a smaller segment, especially among motorized users (9%), but they still constitute 13–14% of the nonmotorized categories. Meanwhile, retired households show an interesting contrast, accounting for 18% of both walk-only and nonmotorized users but just 7% of bike-only users.

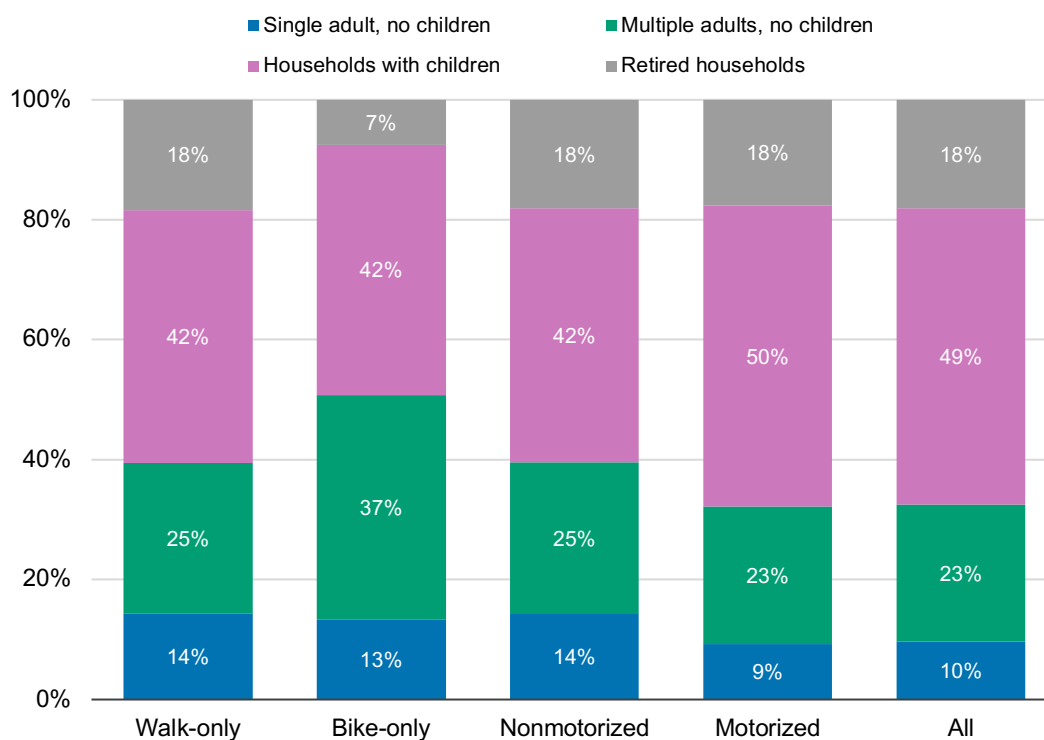


Figure 3-5. Distribution of User Groups by Household Lifecycle in NYS

3.1.6 Household Income

Household income levels could substantially influence travel behavior, including access to transportation modes and the likelihood of engaging in nonmotorized travel. As shown in Figure 3-6, individuals from lower-income households (<\$50,000) are the most prevalent among walk-only and nonmotorized user groups, each comprising 40% of those categories. This trend indicates that walking and biking may serve as more essential travel options among lower-income populations, potentially due to limited access to personal vehicles or travel cost considerations.

In contrast, individuals from the highest income bracket (>\$150,000) show a more balanced distribution across user groups, accounting for 19% of walk-only and nonmotorized users, and similarly 19% of

motorized users. Interestingly, this group also makes up 27% of bike-only users, which may suggest that biking may be more popular among higher-income individuals as a discretionary or recreational mode. The middle-income groups (\$50k–\$100k and \$100k–\$150k) display more varied engagement across modes. Households earning between \$50k and \$100k are well represented among bike-only users (33%) and motorized users (31%). Meanwhile, the \$100k–\$150k group shows greater concentration in motorized travel (20%) and lower representation in nonmotorized categories (13–14%). These findings highlight how nonmotorized travel is often more prominent among lower-income households out of necessity, while biking, in particular, shows a dual pattern: being both a necessity for some and a lifestyle or recreational choice for others in higher-income brackets.

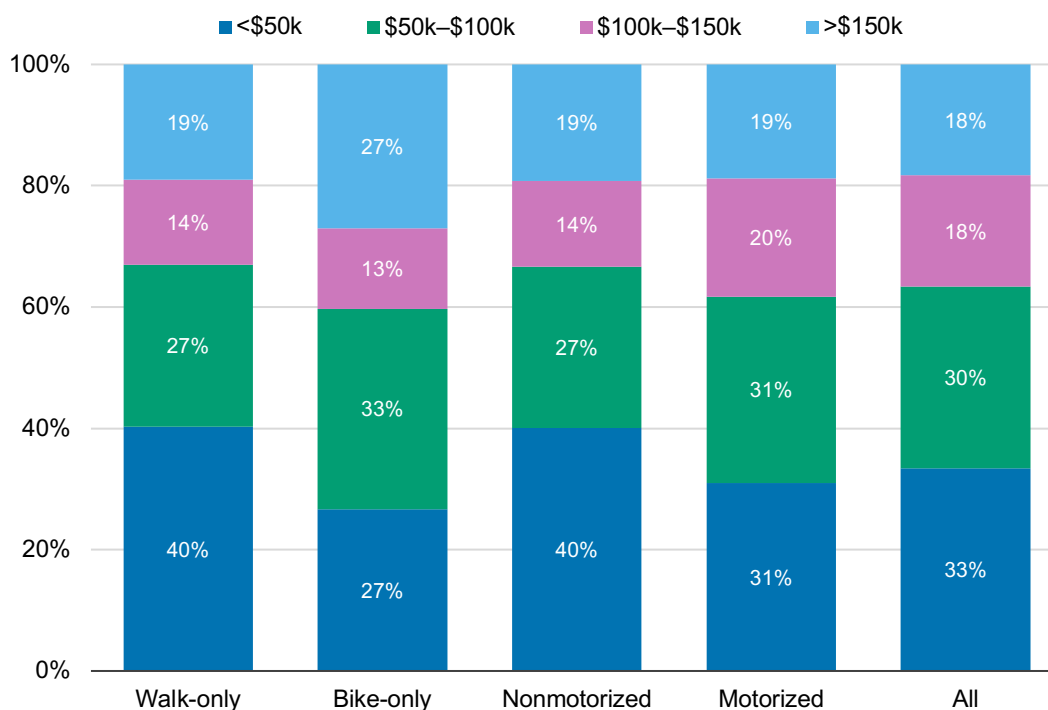


Figure 3-6. Distribution of User Groups by Household Income in NYS

3.1.7 Home Ownership

Homeownership status appears to be historically correlated with travel behavior, particularly in terms of reliance on motorized versus nonmotorized modes. Figure 3-7 below shows that renters make up a larger share of nonmotorized users, with 58% of both walk-only and nonmotorized users residing in rental properties. This pattern is even more pronounced among bike-only users, where 63% are renters. These findings suggest that nonmotorized travel is more common among individuals who rent. This could potentially reflect limited vehicle access, denser housing environments, or proximity to urban centers where walking and biking are more feasible.

In contrast, individuals who own their homes are more dominant among motorized users, accounting for 62% of that group. They also represent a majority (59%) of the population. This difference may reflect the suburban or rural location of homeowners, where car dependency is more prevalent due to lower population density and greater distances to destinations. Overall, the data points to a clear divide in travel patterns based on housing tenure: renters are more likely to engage in nonmotorized travel modes, while homeowners are more likely to rely on motorized transportation.

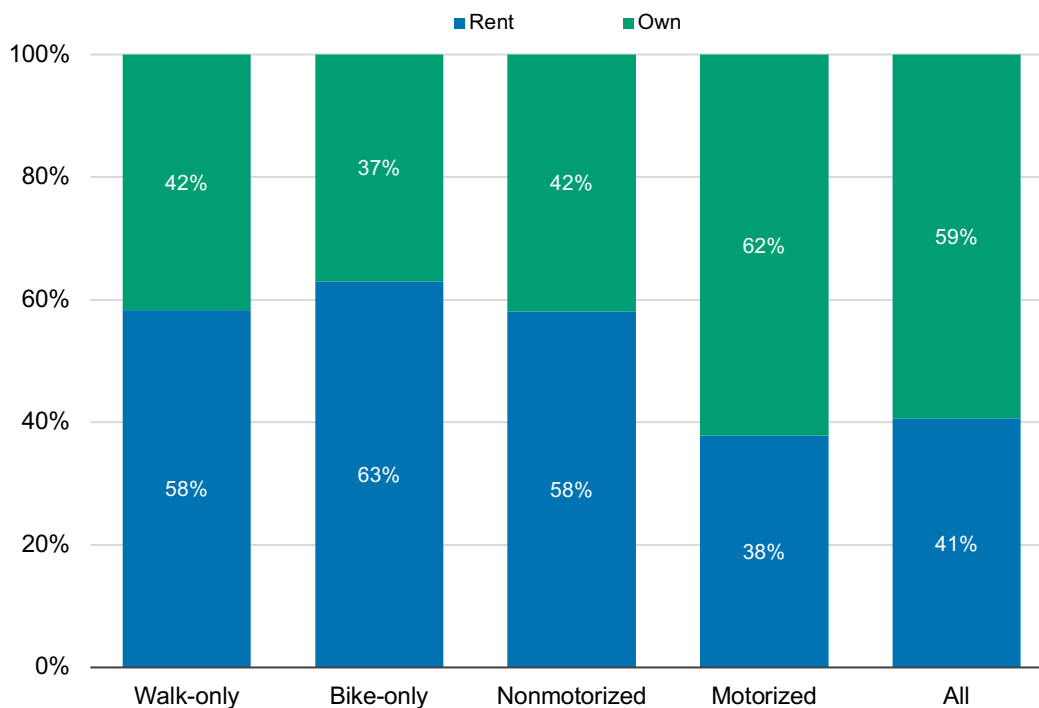


Figure 3-7. Distribution of User Groups by Home Ownership in NYS

3.2 TRIP PATTERNS BY SOCIO-DEMOGRAPHICS AND USER GROUPS

Examining five modal user categories—walk-only, bike-only, nonmotorized (combined walking and biking), motorized, and all-mode users—this analysis compares average daily nonmotorized trip rates across demographic, socioeconomic, and travel characteristics. Trip rates are calculated as person-weighted nonmotorized trips per participating users in respective group. The analysis also leverages the associated margins of error (MOE) to assess whether differences in estimates are statistically significant.

3.2.1 Race

Figure 3-8 presents the average number of daily nonmotorized trips made per person across different racial groups, disaggregated by user group. The racial categories shown include white, black, and others, while user groups are defined based on individuals' travel modes on the survey day: walk-only, bike-only, nonmotorized (any walking and/or biking), motorized (motorized trips only), and all (all trip-makers regardless of mode). Among those in the walk-only group, trip rates are relatively similar across racial groups: white individuals average 2.8 trips, followed by others (2.7) and black individuals (2.6). Although these estimates are very close in magnitude, considering the MOE for each group, ranging from ± 0.03 to ± 0.07 , we find statistically significant differences in walking frequency between white and black individuals, as well as between white and others. For bike-only users, white individuals again show the highest average with 3.7 trips per day, while others and black individuals report slightly lower rates at 3.6 and 3.5, respectively. However, the MOEs are considerably wider in this group. Looking at the broader nonmotorized group, those who made at least one walk or bike trip (with or without motorized travel), white individuals average 2.8 trips per day, slightly more than others (2.7) and black individuals (2.6). Similar to walk-only group, the MOEs in this user group indicate statistically significant differences. In contrast, the motorized group shows somewhat greater variation. Black individuals average 0.7 walk/bike

trips per day, while others and whites average 0.8 and 0.6, respectively. The difference between black and white individuals are significant in this group.

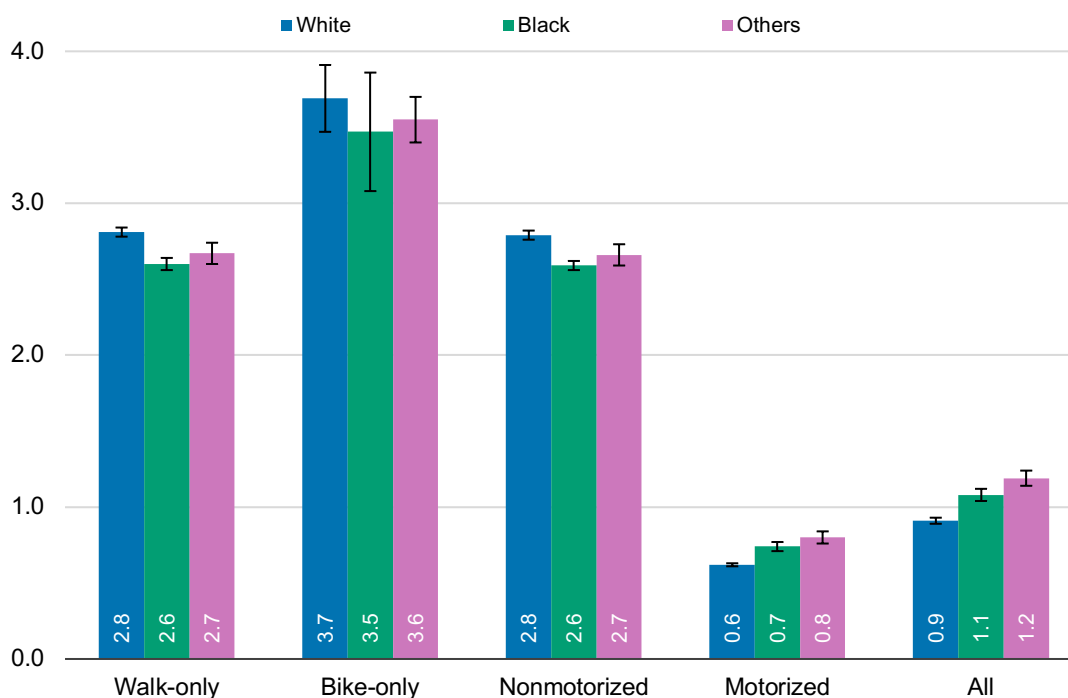


Figure 3-8. Average Daily Nonmotorized Trips by Race and User Group in NYS

3.2.2 Age

Table 3-3 presents the average number of daily nonmotorized trips across five age groups: 5–15, 16–24, 25–39, 40–64, and 65+, broken down by user groups. A clear age gradient is observed in the frequency of nonmotorized travel, with working-age adults generally engaging in more active travel than younger and older populations.

Among walk-only users, the trip rate increases gradually from 2.1 for the 5–15 group to 2.9 for the 25–39 age group. It remains stable at 2.8 for both the 40–64 and 65+ groups, suggesting that walking frequency is consistently higher among adults. The related MOEs provide clear evidence of statistical significance. The bike-only group shows a more distinct peak among middle-aged adults, with trip rates climbing from 2.5 among youth (5–15) to 3.9 and 4.1 for the 25–39 and 40–64 groups, respectively—before dropping to 3.0 for seniors. In the nonmotorized category, which includes anyone taking at least one walk or bike trip, the trend mirrors the walk-only group: lower average trip counts for youth (2.1) and young adults (2.6), peaking at 2.9–2.8 for the 25–64 groups. Seniors (65+) maintain relatively high nonmotorized trip rates at 2.8.

The motorized-only group exhibits a reversed trend. Children (5–15) and seniors (65+) have lower average walk/bike trip counts (0.5 and 0.7), while working-age adults (25–64) average between 0.7 and 0.8 trips per day. When considering all individuals regardless of mode, adults aged 25–39 and 65+ show the highest overall trip rates at 1.1, followed by the 40–64 group at 1.0. The youngest group (5–15) makes the fewest total trips at 0.8.

To better understand how age-based patterns vary geographically, Figure 3-9 compares NYC with the rest of NYS. In NYC, walk-only trip rates are consistently high across all age groups, ranging from 2.3 to 3.1. However, in the rest of NYS, walk-only trip rates are lower across all age groups, ranging between 2.0 and 2.4. This contrast suggests that NYC’s dense, walkable infrastructure enables more consistent walking behavior, even among seniors. A similar disparity exists in bike-only trips. NYC adults aged 25–64 average 4.1–4.4 bike trips daily, far exceeding their counterparts in the rest of NYS, who report 2.9–3.0 trips. The 16–24 group in NYC is missing bike estimates due to omission caused by low sample size, but in the rest of NYS, this group still reports a notable 3.4 average bike trips. For the motorized group, NYC residents average 1.0–1.4 nonmotorized trips, while in the rest of NYS, these values are substantially lower at 0.3–0.4. This could underscore a stronger reliance on walking and biking in NYC compared to more car-dependent environments upstate.

Table 3-3. Average Daily Nonmotorized Trips by Age and User Group

Age group	Estimates					MOE				
	Walk-only	Bike-only	NMT	MT	All	Walk-only	Bike-only	NMT	MT	All
5–15	2.1	2.5	2.1	0.5	0.8	0.04	0.09	0.04	0.01	0.02
16–24	2.7	3.0	2.6	0.5	0.9	0.04	0.23	0.04	0.03	0.05
25–39	2.9	3.9	2.9	0.8	1.1	0.04	0.18	0.04	0.02	0.02
40–64	2.8	4.1	2.8	0.7	1.0	0.03	0.19	0.03	0.01	0.02
65+	2.8	3.0	2.8	0.7	1.1	0.06	0.11	0.06	0.03	0.03

NMT = Nonmotorized, MT = Motorized, MOE = Margin of Error

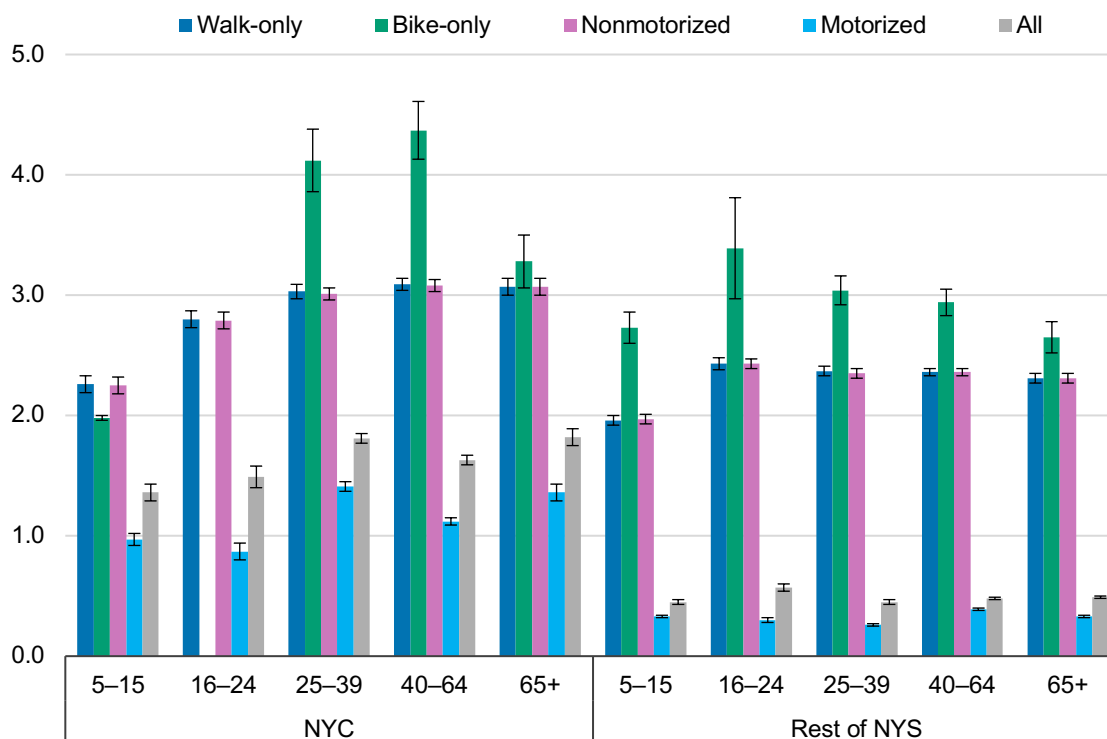


Figure 3-9. Average Daily Nonmotorized Trips by Age and User Group in NYC and the Rest of NYS

The generational distribution presented in Figure 3-10 reveals distinct patterns in nonmotorized travel behavior. Gen X made significantly more walk-only trips compared to other older and younger generations. This is true for the overall nonmotorized group. However, for bike-only group, there is no statistical difference between Gen X and Baby Boomers, although they made significantly more bike trips than Millennials, Gen Z and other older generations.

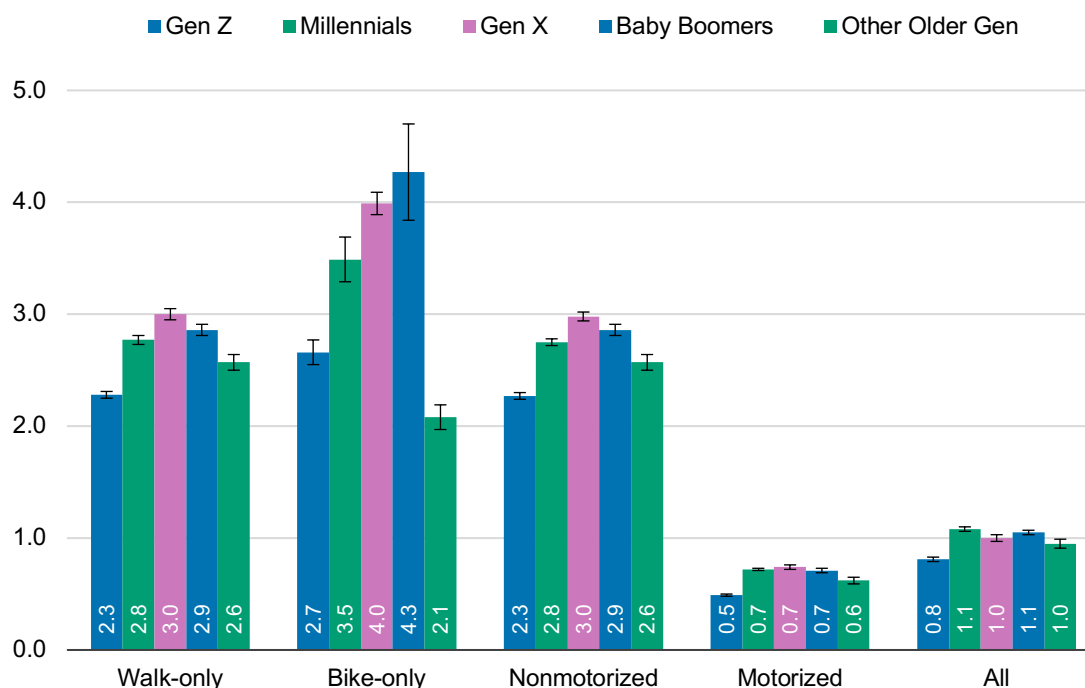


Figure 3-10. Average Daily Nonmotorized Trips by Generation and User Group in NYS

3.2.3 Sex

Table 3-4 presents the nonmotorized trips per person by sex and user group. In the walk-only group, females average 2.7 daily walk trips, while males report a slightly higher rate of 2.8. However, the difference is not statistically significant. The bike-only category reveals a more noticeable sex difference: females report an average of 3.9 bike trips compared to 3.5 for males. Among those who took any nonmotorized trip, both sexes show very similar results: 2.7 trips per day for females and 2.8 for males. However, this difference is not statistically significant. In summary, while females exhibit slightly higher average bike trips, and males walk marginally more, none of these differences are statistically significant. This suggests that, overall, nonmotorized travel behavior in NYS is relatively balanced across sexes.

Table 3-4. Average Daily Nonmotorized Trips by Sex and User Group in NYS

Sex	Estimates					MOE				
	Walk-only	Bike-only	NMT	MT	All	Walk-only	Bike-only	NMT	MT	All
Female	2.7	3.9	2.7	0.7	1.1	0.02	0.14	0.02	0.01	0.01
Male	2.8	3.5	2.8	0.6	1.0	0.03	0.15	0.03	0.02	0.02

NMT = Nonmotorized, MT = Motorized, MOE = Margin of Error

3.2.4 Household Size

Figure 3-11 illustrates the average number of daily nonmotorized trips per person across households of varying sizes, segmented into three groups: one-person households, two-person households, and households with more than two people. Within the walk-only group, individuals from one-person households average 3.1 trips per day, which is the highest among all household sizes. This is followed by two-person households with 2.8 trips, and more-than-two-person households with 2.5 trips. The difference between one-person and larger households is statistically significant. In the bike-only group, a similar trend is observed. Two-person households show the highest average at 4.5 trips per day, slightly exceeding one-person households at 4.3 trips. However, individuals from larger households report a significantly lower average of just 2.9 trips.

Looking at the overall nonmotorized group, individuals from one- and two-person households average 3.1 and 2.8 trips respectively, whereas those in larger households average only 2.5 trips. These differences are statistically significant. In the motorized-only group, walk and bike trip frequency decreases with household size. One-person households average 1.2 motorized trips, followed by two-person households at 0.8, and larger households at just 0.5. The lower mobility observed in larger households may be attributable to constraints like childcare responsibilities, resource sharing, or reliance on fewer members to complete travel for the household. Overall, this analysis shows that smaller households, particularly one- and two-person households, are more likely to engage in higher levels of nonmotorized travel, while larger households demonstrate significantly lower trip frequencies.

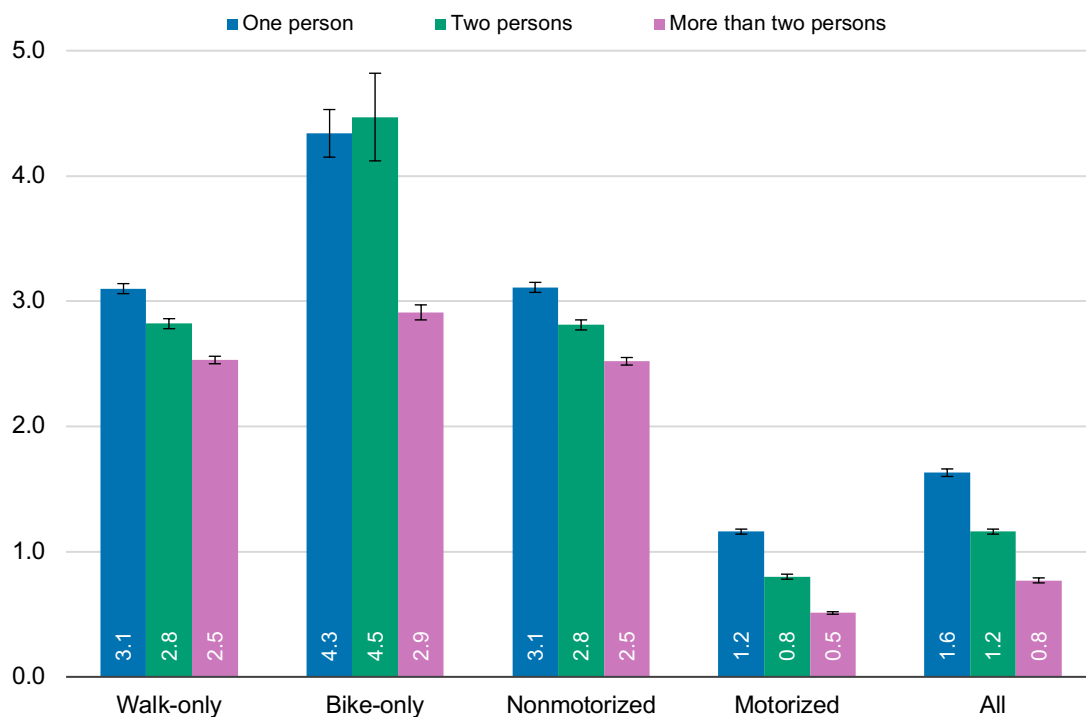


Figure 3-11. Average Daily Nonmotorized Trips by Household Size and User Group in NYS

3.2.5 Household Income

Figure 3-12 presents the average number of daily trips across four income brackets: <\$50k, \$50k–\$100k, \$100k–\$150k, and >\$150k. Across all income groups, bike-only trips are the highest across the regions.

NYC bike-only trips are higher than the rest of NYS. Specifically, in NYC, individuals from households earning less than \$50k averaged 3.0 in walk-only and 5.5 trips per day in bike-only group. These translate to 3.0 total trips in nonmotorized user group. Notably, these differences are statistically significant.

In contrast, middle-income groups (\$50k–\$150k) maintain slightly lower but relatively stable nonmotorized rates (2.7–3.0), while the highest-income group (> \$150k) again reports high walk-only (3.1) and bike-only (4.3) trips. This group also records the highest overall trip average (1.9). These findings indicate a bimodal distribution wherein nonmotorized travel demonstrates high prevalence across income categories within NYC, with particular concentration among both low-income and high-income households.

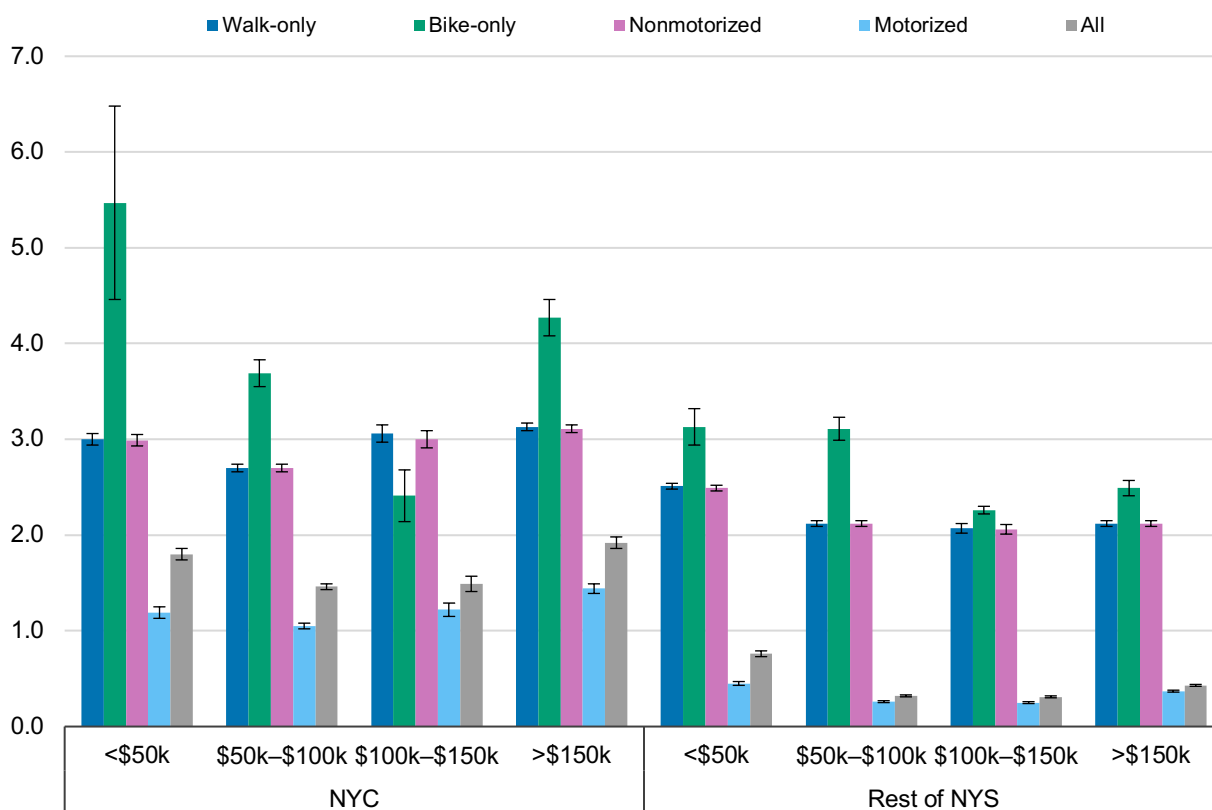


Figure 3-12. Average Daily Nonmotorized Trips by Household Income and User Group in NYC and the Rest of NYS

Going through the rest of NYS, the pattern is somewhat linear: as income increases, nonmotorized travel rates steadily decline. Individuals earning less than \$50k average 2.5 walk-only and 3.1 bike-only trips per day (2.5 total nonmotorized trips), while those in the >\$150k group average just 2.1 walk-only, 2.5 bike-only, and 2.1 total nonmotorized trips. This statistically significant trend suggests the existence of income-based disparities in less urbanized settings.

3.2.6 Household Lifecycle

For walk-only travel, NYC shows the clearest gradient (Table 3-5): single adults average 3.3 trips and retired households 3.2, both significantly above multiple-adult households (2.9 trips) and households with children (2.7 trips). Single vs. retired overlap slightly, indicating no statistical difference between those

two groups. In the rest of NYS, single adults also exceeded other lifecycles in walk trips: their MOE range does not overlap with multiple-adult (2.3), households with children (2.3), or retired (2.2), confirming a statistically higher walking rate for singles.

The bike-only pattern differs by region. In NYC, single adults ride most (5.2), significantly above multiple-adult (4.3), households with children (3.2), and retired households (2.6). In the rest of NYS, the peak shifts to multiple-adult, no-children households (3.4), which is significantly higher than singles (3.1) and households with children (2.8). Retired households (2.9) sit between those groups; their MOE band overlaps with households with children, so that difference is not statistically significant. Looking at the broader nonmotorized category (any walk and/or bike), NYC single adults again lead, followed by retired; both are significantly higher than multiple-adult and households with children. The Rest of NYS resonates this ranking at a lower level. For instance, single adults exceed multiple-adult, child, and retired households, with non-overlapping MOEs. Overall, across regions, single-adult households exhibit the highest nonmotorized trip rates, with NYC retirees also making nonmotorized trips near single-adult household levels. Households with children consistently show the lowest nonmotorized activity, and outside NYC, multiple-adult households are the most active cyclists.

Table 3-5. Average Daily Nonmotorized Trips by Household Lifecycle and User Group

Region	Lifecycle	Estimate					MOE				
		Walk-only	Bike-only	NMT	MT	All	Walk-only	Bike-only	NMT	MT	All
NYC	Single adult, no children	3.3	5.2	3.3	1.6	2.2	0.05	0.25	0.05	0.05	0.06
	Multiple adults, no children	2.9	4.3	2.8	1.2	1.6	0.05	0.41	0.05	0.04	0.04
	Households with children	2.7	3.2	2.7	1.1	1.5	0.06	0.1	0.06	0.05	0.06
	Retired households	3.2	2.6	3.1	1.2	1.8	0.06	0.25	0.06	0.07	0.07
Rest of NYS	Single adult, no children	3.3	5.2	3.3	1.6	2.2	0.05	0.25	0.05	0.05	0.06
	Multiple adults, no children	2.9	4.3	2.8	1.2	1.6	0.05	0.41	0.05	0.04	0.04
	Households with children	2.7	3.2	2.7	1.1	1.5	0.06	0.1	0.06	0.05	0.06
	Retired households	3.2	2.6	3.1	1.2	1.8	0.06	0.25	0.06	0.07	0.07

NMT = Nonmotorized, MT = Motorized, MOE = Margin of Error

3.2.7 Vehicle Ownership

In Figure 3-13, we compare trip-making patterns across different levels of household vehicle ownership within each user group, separately for NYC and the rest of NYS. The vehicle ownership categories include zero vehicle, one-vehicle, two-vehicle, and more than two-vehicle households.

For walk-only travelers, the pattern is intuitive: households without a vehicle consistently record the highest number of walking trips, both in NYC and the rest of NYS. As the number of household vehicles increases, walk-only activity tends to decline. This drop is especially pronounced in the rest of NYS,

where households with two or more vehicles show the lowest levels of walk-only trips. In NYC, although households with more than two vehicles appear to have high walking activity; however, the differences are statistically not significant. Nonmotorized trips show similar trends. Zero vehicle households are the most active, followed by those with one vehicle, while households with two or more vehicles tend to engage less in nonmotorized travel. These patterns are largely consistent and statistically consistent across both NYC and the rest of NYS.

Bike-only travel shows a slightly more nuanced pattern. In NYC, bike usage is higher among households with zero or one vehicle compared to those with two vehicles. However, estimates for the more than two-vehicle households category carry wide MOEs. In the rest of NYS, bike-only activity exhibits a steady decline with increasing vehicle ownership. Finally, looking at motorized-only users, households without a vehicle report a notable number of nonmotorized trips in NYC, which is higher than other groups. In contrast, in the rest of NYS, nonmotorized travel by motorized-only users decreases with greater vehicle ownership. Overall, the relationship between vehicle ownership and mode choice is clear: fewer household vehicles are associated with greater reliance on nonmotorized options. These differences are both behaviorally intuitive and statistically significant in most user groups.

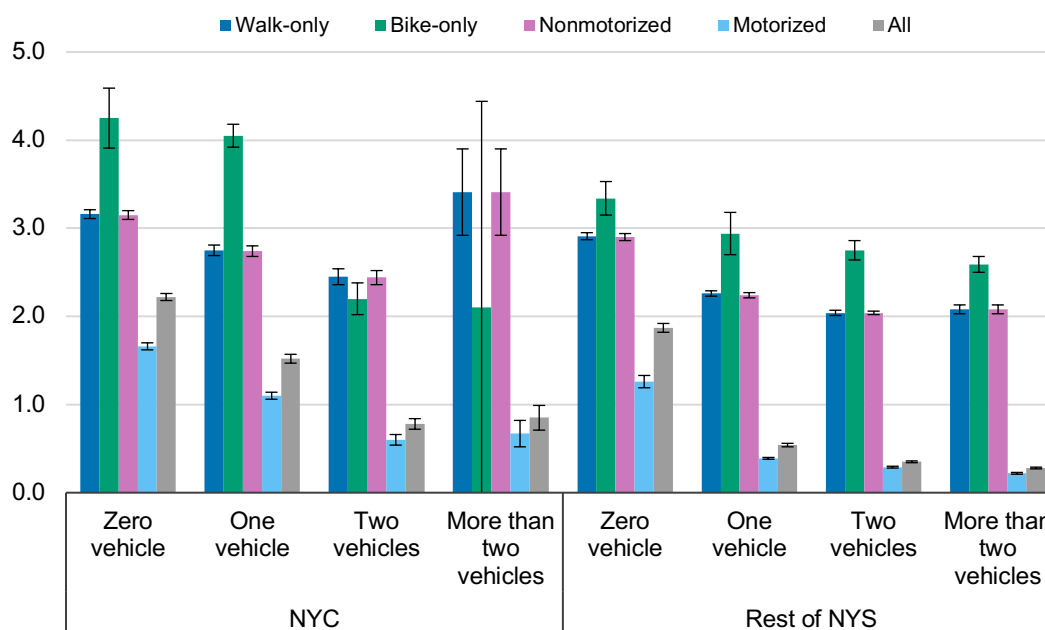


Figure 3-13. Average Daily Nonmotorized Trips by Household Vehicle Ownership by User Group in NYC and the Rest of NYS

3.2.8 Driver Availability

Table 3-6 shows how nonmotorized trip-making varies across households with differing numbers of licensed drivers in NYC and the rest of NYS. The number of drivers can reflect both individual mobility capacity and broader household-level travel dynamics. In NYC, households with one or two drivers exhibit similarly high rates of walk-only and nonmotorized trips. These groups also show elevated bike-only usage, particularly among two-driver households. However, when we look at households with more than two drivers, a noticeable drop in nonmotorized travel occurs. This group engages in significantly fewer walking and biking trips. This could indicate the presence of a larger pool of drivers, which allows greater reliance on car travel, even among those typically categorized under motorized or all-user groups.

The pattern is even more pronounced in the rest of NYS. Here, one-driver households lead in walk-only and overall nonmotorized activity. As the number of drivers increases, walking and biking rates steadily decline. Two-driver and more-than-two-driver households display reduced engagement in nonmotorized travel, suggesting that driver availability enables greater auto dependency. Among users classified under motorized and all categories, their nonmotorized trip levels similarly taper off as the number of drivers increases. This trend indicates that even within groups that depend on cars, the presence of fewer drivers is associated with a higher likelihood of incorporating walking or biking into daily routines.

Table 3-6. Average Daily Nonmotorized Trips by Number of Drivers in Households by User Group in NYC and the Rest of NYS

Region	Drivers	Estimate					MOE				
		Walk-only	Bike-only	NMT	MT	All	Walk-only	Bike-only	NMT	MT	All
NYC	One driver	3.0	3.6	3.0	1.3	1.8	0.06	0.14	0.06	0.05	0.04
	Two drivers	3.0	4.7	3.0	1.2	1.6	0.06	0.4	0.06	0.05	0.05
	More than two drivers	2.4	2.2	2.4	0.6	0.9	0.12	0.13	0.12	0.07	0.06
Rest of NYS	One driver	2.5	2.7	2.4	0.5	0.7	0.04	0.28	0.03	0.02	0.03
	Two drivers	2.1	2.9	2.1	0.3	0.4	0.02	0.1	0.02	0.01	0.01
	More than two drivers	2.1	2.6	2.1	0.2	0.3	0.06	0.1	0.05	0.01	0.01

NMT = Nonmotorized, MT = Motorized, MOE = Margin of Error

3.2.9 Travel Time

Figure 3-14 summarizes the trend across four travel time groups: ≤10 minutes, 10–30 minutes, 30–60 minutes, and > 60 minutes. For walk-only users, the pattern is quite clear. Those with shorter average travel times (under 30 minutes) tend to walk more frequently. The highest walking trip rates are seen among those traveling 10 minutes or less and those in the 10–30 minutes group. Beyond 30 minutes, walking activity begins to drop significantly, especially for those whose travel takes over an hour.

Bike-only users show a slightly different trend. While biking remains consistent across the first three groups (up to 60 minutes), the most frequent riders are actually in the 30–60 minute range. However, a significant decline in biking activity occurs when travel time exceeds 60 minutes, indicating that longer overall travel durations might discourage daily biking. When combining walk and bike travel into the nonmotorized category, the trend generally follows that of walk-only users. Shorter travel times align with higher nonmotorized activity, with diminishing returns as travel time increases. Those with the shortest commutes report the most nonmotorized trips, while the group traveling over an hour daily shows the lowest levels of such activity.

Even among motorized-only users, we see that those with shorter travel times make more nonmotorized trips. This further reinforces the idea that even individuals who rely on motorized modes still supplement their travel with walk or bike trips. Overall, the data suggests a clear inverse relationship between average travel time and nonmotorized travel activity. As total daily travel time increases, the number of walking and biking trips per person tends to decrease. Shorter travel time seems to enable and possibly encourage greater engagement in nonmotorized travel across all groups.

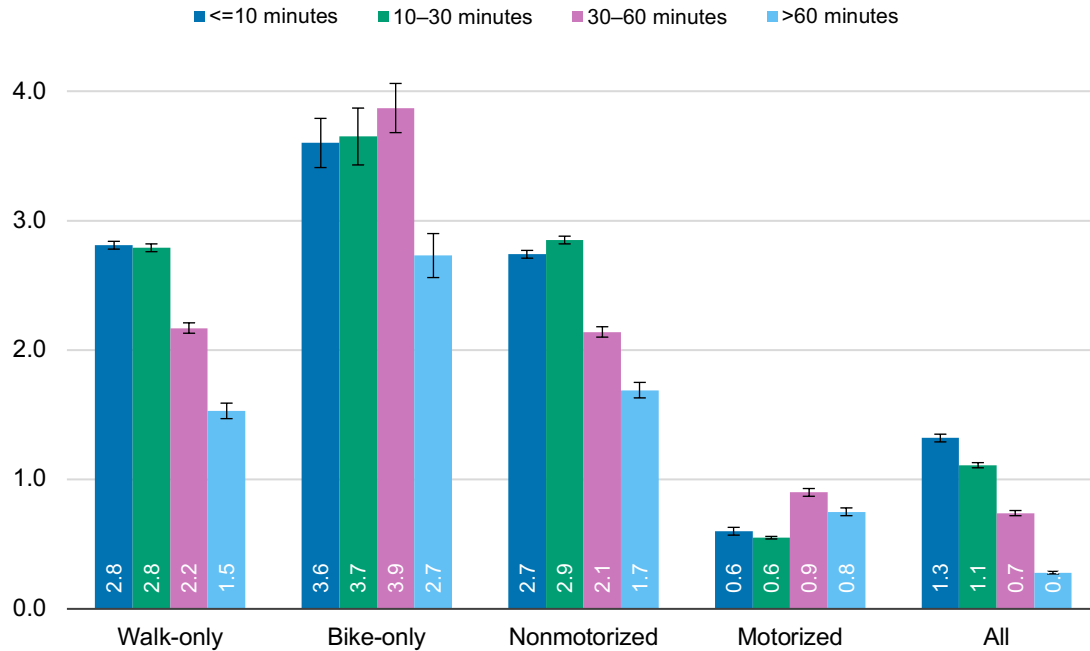


Figure 3-14. Average Daily Nonmotorized Trips by Average Travel Time by User Group in NYS

3.2.10 Rideshare

How participation in ridesharing services influences average daily nonmotorized trips across different user groups and regions is explored (Figure 3-15). The comparison is made between individuals who reported using rideshare services (such as Uber or Lyft) and those who did not, within both NYC and the rest of NYS.

In NYC, nonmotorized trip-making appears slightly higher among rideshare users. Walk-only users who participate in rideshare services average 3.1 daily walk trips compared to 2.9 among non-users. While the difference is minimal, it is statistically significant. Similarly, nonmotorized trip rates are higher for rideshare users (3.1 vs. 2.9), suggesting a complementary—not substitutive—relationship between walking and rideshare in NYC. Interestingly, the bike-only rates are nearly identical between users and non-users of rideshare (3.9 vs. 4.0). Additionally, those in the motorized-only group who also use rideshare exhibit much higher nonmotorized trip rates (1.6 vs. 1.1), potentially reflecting multimodal lifestyles where rideshare fills mobility gaps rather than fully replacing active travel. Altogether, NYC data suggests that rideshare participation correlates with slightly higher nonmotorized mobility across all user groups.

In contrast, residents of the rest of NYS show smaller and more ambiguous differences. Walk-only and bike-only trip rates are nearly identical between rideshare users and non-users. Walk-only users show a negligible increase (2.4 vs. 2.3), while bike-only users record a lower value among rideshare participants (2.4 vs. 3.0). The nonmotorized group average remains flat at 2.4 for users and 2.3 for non-users, which is not statistically significant. Motorized-only users again show an uptick in nonmotorized trip-making if they are also ridesharing users (0.5 vs. 0.3). Overall, the rest of NYS displays weaker and more inconsistent relationships between rideshare use and active travel.

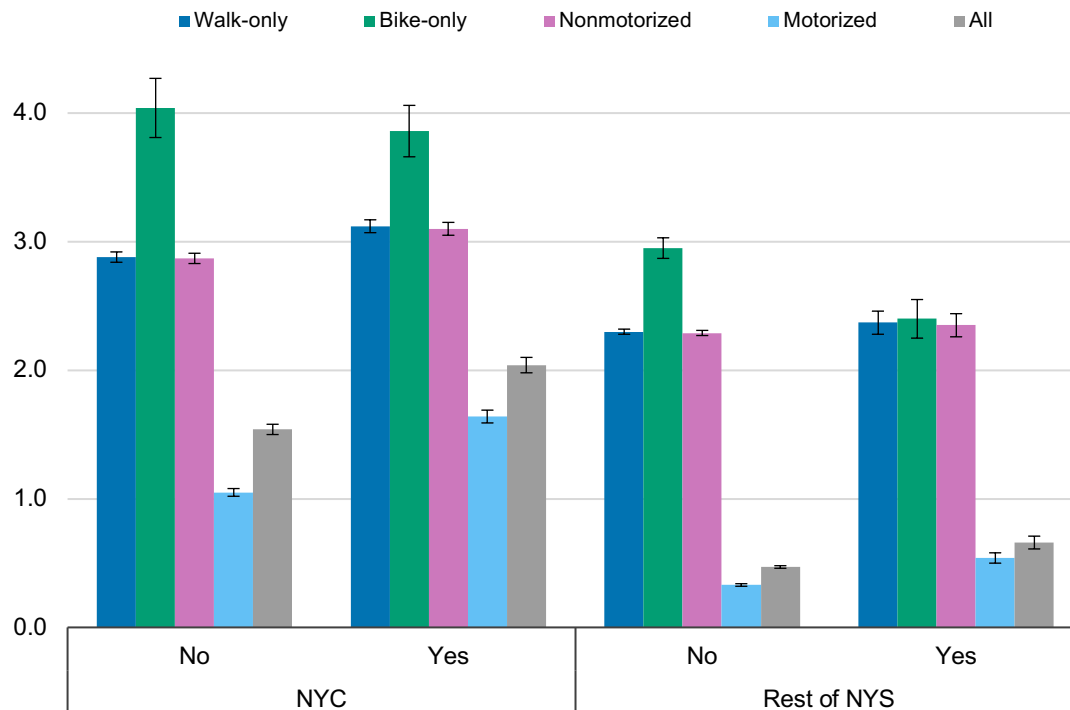


Figure 3-15. Average Daily Nonmotorized Trips by Rideshare Participation by User Group in NYC and the Rest of NYS

3.2.11 Carshare

Carshare users in NYC demonstrate significantly higher levels of nonmotorized travel than their non-carshare counterparts (Figure 3-16). The most notable difference is seen in bike-only trips, where carshare users average 6.9 trips daily, compared to just 3.7 trips for non-users. Walk-only trips also show a positive association with carshare use in NYC. We can see 3.5 daily trips among carshare users versus 2.9 for non-users. Similarly, the overall nonmotorized trip rate is 3.4 for users compared to 2.9 for non-users. These patterns suggest that carshare participants in NYC are more likely to combine shared motorized modes with active travel options. This could be due to reduced reliance on personal vehicles and better integration of carshare with walking and biking infrastructure. In the rest of NYS, the same general trend is visible, though at lower levels. Carshare participants recorded higher walk-only (2.9 vs. 2.3), bike-only (3.4 vs. 2.9), and nonmotorized (2.9 vs. 2.3) trip rates compared to non-users.

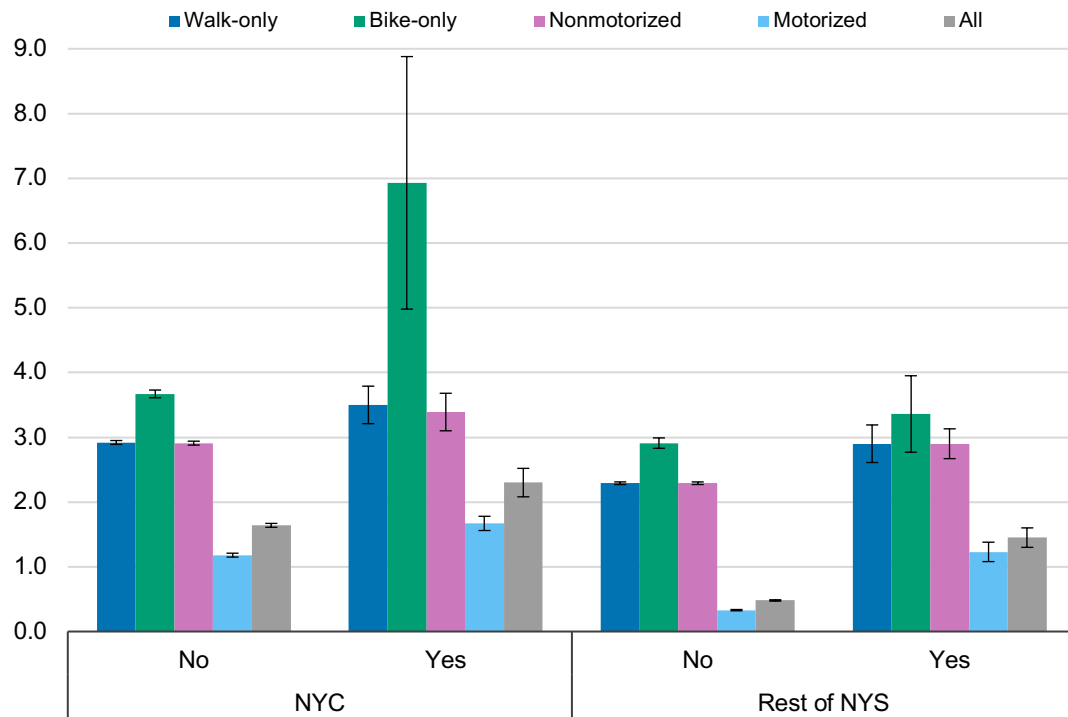


Figure 3-16. Average Daily Nonmotorized Trips by Carshare Usage by User Group in NYC and the Rest of NYS

4. NONMOTORIZED TRAVEL BY HOUSEHOLD ALICE STATUS

This section explores how nonmotorized travel behavior differs between ALICE and non-ALICE households across NYS. The analysis will further investigate how these differences manifest across key demographic, socioeconomic, and travel characteristics and within different user groups.

4.1 TRAVEL BY ALICE STATUS

Figure 4-1 shows regional patterns in the share of nonmotorized trips among ALICE and non-ALICE households. Across both NYC and the rest of NYS, ALICE households consistently exhibit a higher reliance on nonmotorized travel compared to their non-ALICE counterparts. In NYC, nonmotorized trips make up 48% of all trips taken by ALICE households, compared to 40% for non-ALICE households. This gap could suggest that even within a dense, transit-rich environment, ALICE households may be more dependent on walking or biking, potentially due to lower access to private vehicles or the need to minimize transportation expenses. The contrast becomes even more apparent in the rest of NYS, where nonmotorized trips account for 20% of total trips for ALICE households, but only 9% for non-ALICE households. This more than twofold difference underscores a clear reliance on nonmotorized modes among ALICE households even in less urbanized, possibly less walkable areas.

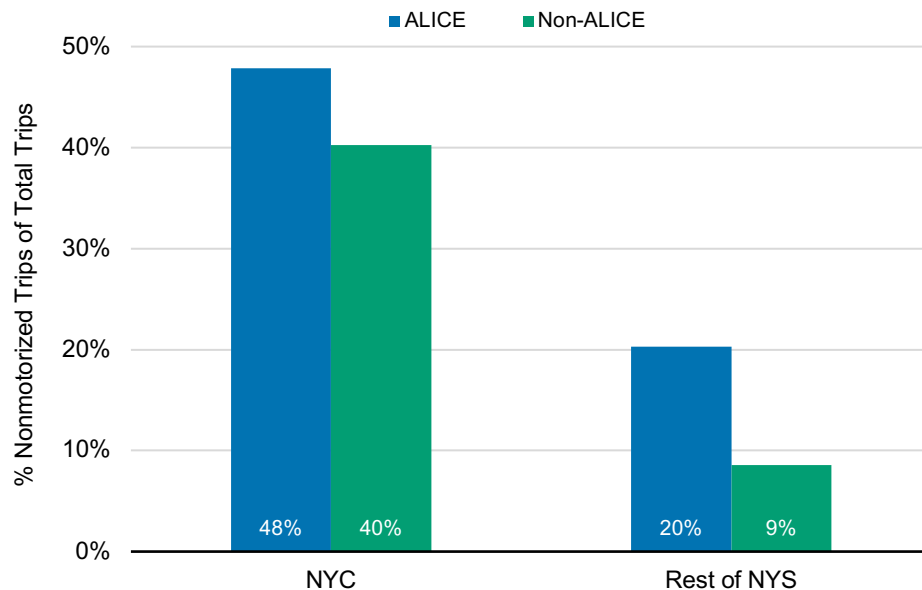


Figure 4-1. Share of Nonmotorized Trips by ALICE Status in NYC and the Rest of NYS

If we look at the distance covered through nonmotorized travel, some notable differences emerge (Figure 4-2). In NYC, ALICE households have a slightly higher share of nonmotorized trip mileage compared to non-ALICE households. While the difference is modest, it suggests that ALICE households in NYC are not only making a larger share of their trips via walking or biking (as seen in the previous figure) but are also traveling somewhat longer distances using these modes. However, the disparity becomes far more pronounced outside of NYC. In the rest of NYS, nonmotorized travel comprises nearly 10% of total mileage for ALICE households, compared to just 2% for non-ALICE households. This gap indicates that ALICE households in non-urban areas are traveling significantly longer distances on foot or by bike.

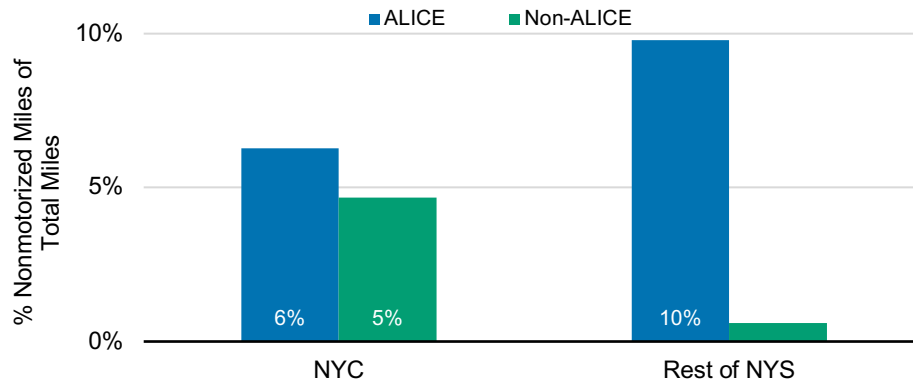


Figure 4-2. Share of Nonmotorized Trip Mileage by ALICE Status in NYC and the Rest of NYS

To further understand the percentage of nonmotorized trips by ALICE households, the role of household vehicle ownership is examined in Figure 4-3. The most apparent observation is the very high level of nonmotorized travel among households with zero vehicles, where both ALICE and non-ALICE households report that 56% of their trips are made by nonmotorized modes. This share suggests that vehicle unavailability is the biggest catalyst when it comes to reliance on walking or biking for daily travel, regardless of economic status. However, once a household has access to a vehicle, differences begin to emerge. Among one vehicle households, non-ALICE households report a higher share of nonmotorized trips (29%) compared to ALICE households (22%). In households with two vehicles, the share of nonmotorized trips drops to 11% for both groups. For households with three or more vehicles, the decline is more notable for non-ALICE households (8%) compared to ALICE households (11%).

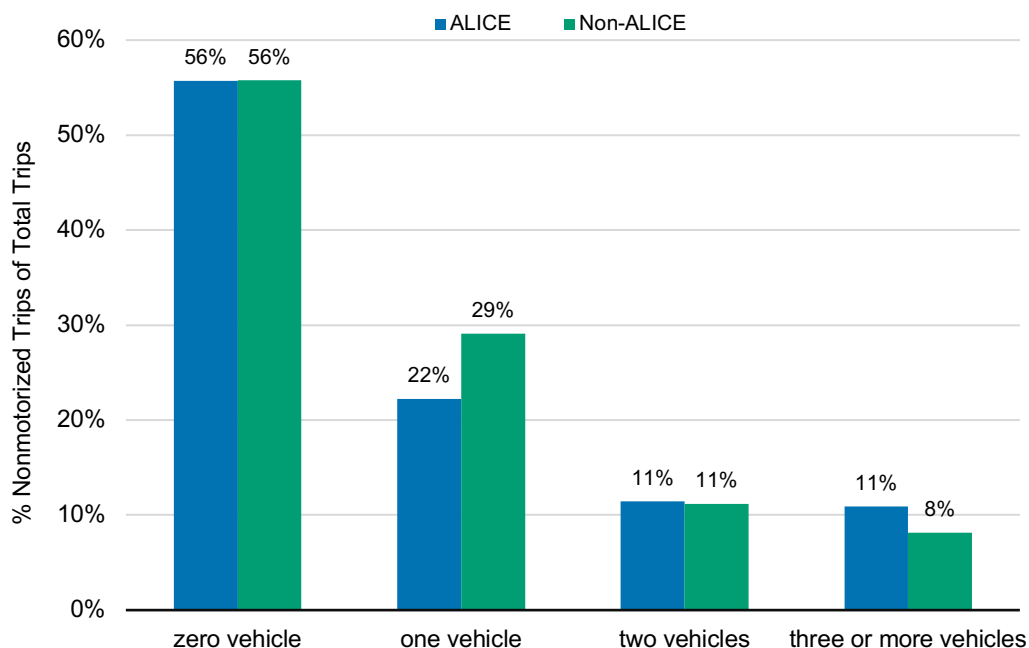


Figure 4-3. Share of Nonmotorized Trips by Household Vehicle Ownership and ALICE Status in NYS

Figure 4-4 breaks down nonmotorized travel by ALICE status across different household vehicle levels separately for NYC and the rest of NYS. For zero vehicle households, nonmotorized travel constitutes the majority of trips in both NYC and the rest of NYS, with shares exceeding 50% for ALICE and non-ALICE households. This underscores the critical role of walking and biking as default travel modes when vehicle access is absent, particularly in high-density areas like NYC. However, in the rest of NYS, non-ALICE households have substantially lower nonmotorized trip share (36%) compared to ALICE households (54%). Among one-vehicle households, the difference between NYC and the rest of NYS becomes pronounced. ALICE households in NYC make 37% of their trips via nonmotorized modes, slightly lower than non-ALICE households (39%). However, outside NYC, ALICE households show a notable drop to 14%, while non-ALICE one-vehicle households drop even further to 12%. For households with more vehicles, nonmotorized mode shares drop further.

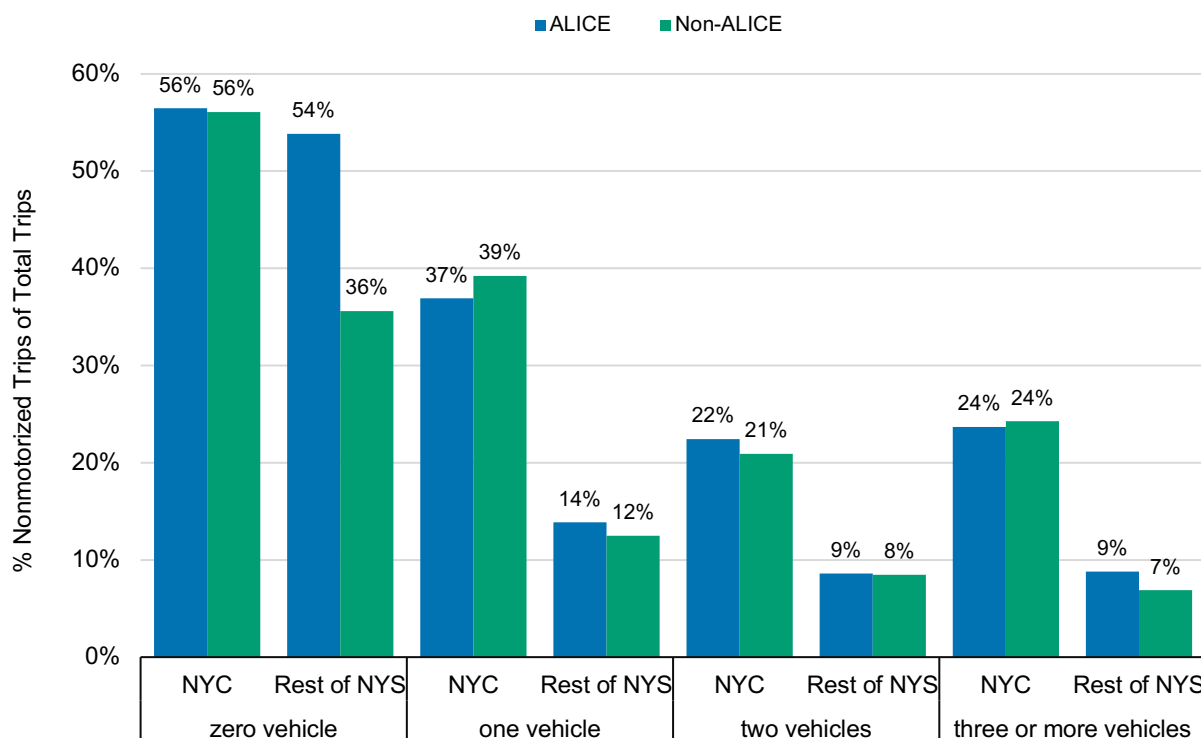


Figure 4-4. Share of Nonmotorized Trips by Household Vehicle Ownership and ALICE Status in NYC and the Rest of NYS

4.2 TRIPS BY TIME OF DAY

Figure 4-5 provides insight into how nonmotorized travel varies by time of day for ALICE and non-ALICE households in NYS. Overall, both groups exhibit a similar temporal distribution of nonmotorized trips throughout the day, but subtle differences in intensity and peak times are noteworthy. During the early morning hours (before 6 a.m.), both groups show negligible levels of nonmotorized activity, which is expected due to limited daylight and travel needs at that time. Starting at 7 a.m., a sharp increase in nonmotorized travel is observed for both groups, with ALICE households showing slightly higher percentages through the morning rush hours (8–10 a.m.).

Interestingly, the non-ALICE group overtakes ALICE in nonmotorized travel share during the midday peak (around 12–1 p.m.), which could reflect discretionary activities such as errands or leisure trips among higher-income households. After the midday peak, both groups again converge in their nonmotorized travel patterns. In the late afternoon and evening hours (3–8 p.m.), both ALICE and non-ALICE households display a gradual decline in nonmotorized trips.

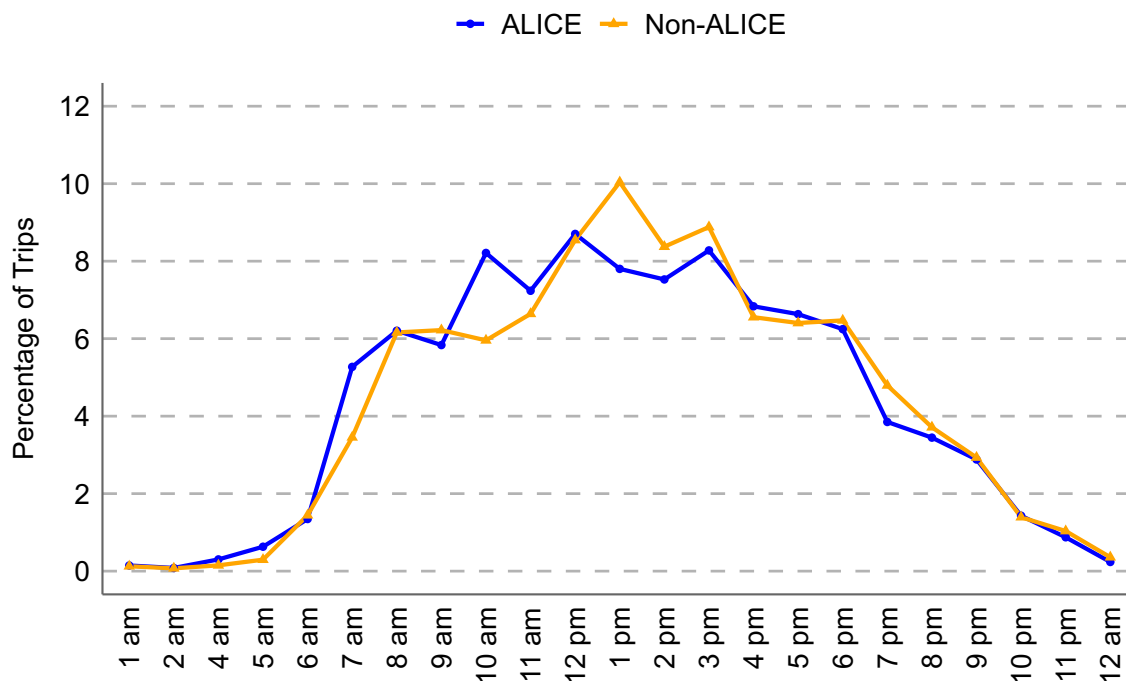


Figure 4-5. Share of Nonmotorized Trips by Time of Day and ALICE Status in NYS

4.3 TRIPS BY TIME OF DAY AND PURPOSE

Figure 4-6 presents a heatmap comparison of when nonmotorized trips occur throughout the day for different trip purposes, separately for ALICE and non-ALICE households across NYS. Each darker cell reflects a higher concentration of trips starting during that hour within that purpose and group, not a higher overall trip count. A clear pattern emerges for *School or Church* purposes, where both ALICE and non-ALICE households show a high concentration of trips in the early morning hours (around 7 a.m.). This aligns with expected school start times and morning drop-offs. However, the peak is more pronounced for ALICE households. Regarding *Family and Personal* nonmotorized trips, non-ALICE households show a notable midday spike (1 p.m. and 4 p.m.). ALICE households also make such trips during midday, but the distribution is more spread out across late morning and early afternoon hours, with less concentration. *Work-related* travel shows similar patterns across both groups. There is more concentration of work-related trips in the morning and evening peaks. Trips related to *Shopping and Recreation* display a broader spread across the day for both groups, with moderate concentrations from late morning through early evening (10 a.m.–6 p.m.). The patterns are relatively similar, though ALICE households exhibit slightly less midday intensity.

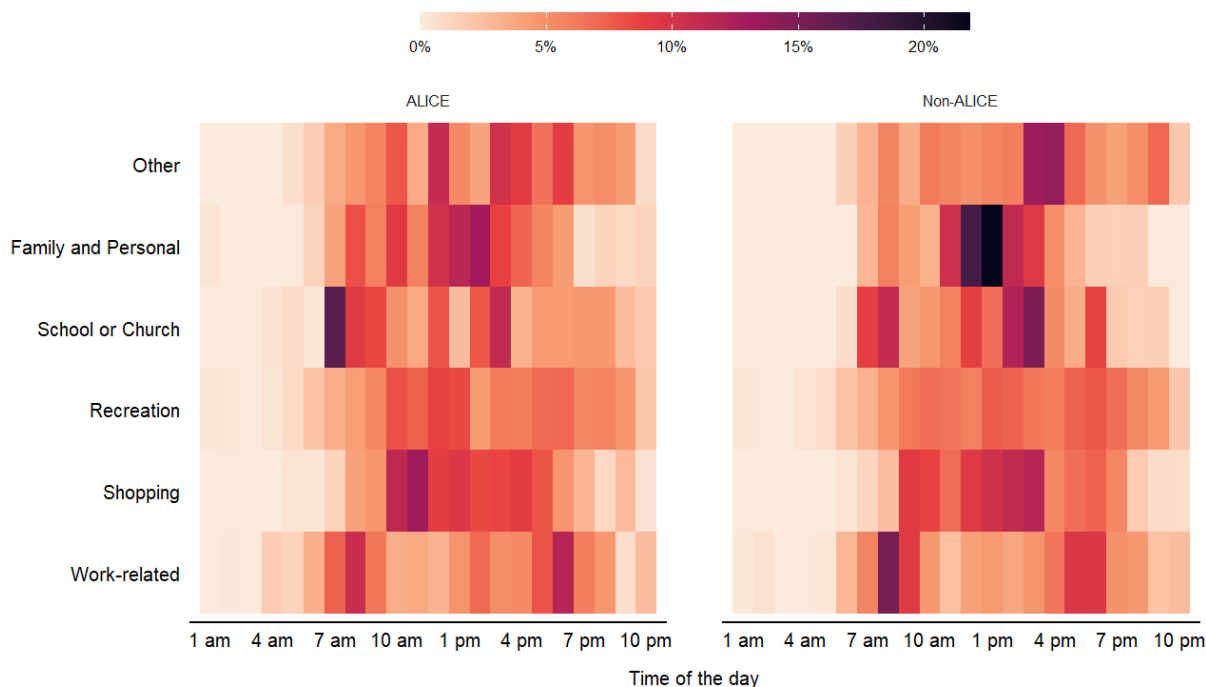


Figure 4-6. Share of Nonmotorized Trips by Hour, Trip Purpose and ALICE Status in NYS

4.4 TRIPS BY PURPOSE AND DAY OF WEEK

In Figure 4-7, we explore how nonmotorized trip purposes vary across weekdays and weekends for ALICE and non-ALICE households in NYS. This highlights several noteworthy trends. On weekdays, *work-related* nonmotorized trips are more prevalent among non-ALICE households (13%) than ALICE households (9%). Conversely, ALICE households make a larger share of weekday nonmotorized trips for *School or Church* (14%) compared to non-ALICE households (7%). *Shopping-related* trips form a significant portion for ALICE households during weekends (31%), which is notably higher than non-ALICE households (19%). On the other hand, *Recreation-related* trips are dominant among non-ALICE households, especially on weekends compared to that for ALICE. This large gap could reflect differences in leisure time availability, access to recreational facilities, or lifestyle preferences. *Family and Personal* purposes see a moderate share across all groups and days, but slightly higher for non-ALICE households on weekdays (22%).

4.5 TRIP CHARACTERISTICS BY USER GROUPS

Figure 4-8 provides a comparative look at nonmotorized trip rates per person, stratified across distinct user groups: walk-only, bike-only, nonmotorized (combined walk and bike), motorized, and all. It is important to note that this figure only reflects nonmotorized trips made by individuals within each group, regardless of whether they also made motorized trips. Walk-only trips are similar across groups (2.8 for ALICE vs. 2.7 for non-ALICE), showing consistent reliance on walking regardless of economic status. When aggregating all nonmotorized modes, ALICE individuals take slightly more trips (2.8 vs. 2.6). For motorized trips, ALICE individuals average more trips (0.8) compared to non-ALICE (0.6). Overall, the total nonmotorized trip rate for all user group is higher for ALICE (1.3) than non-ALICE (0.8). Note that the differences between ALICE and non-ALICE are statistically significant in all the user groups.

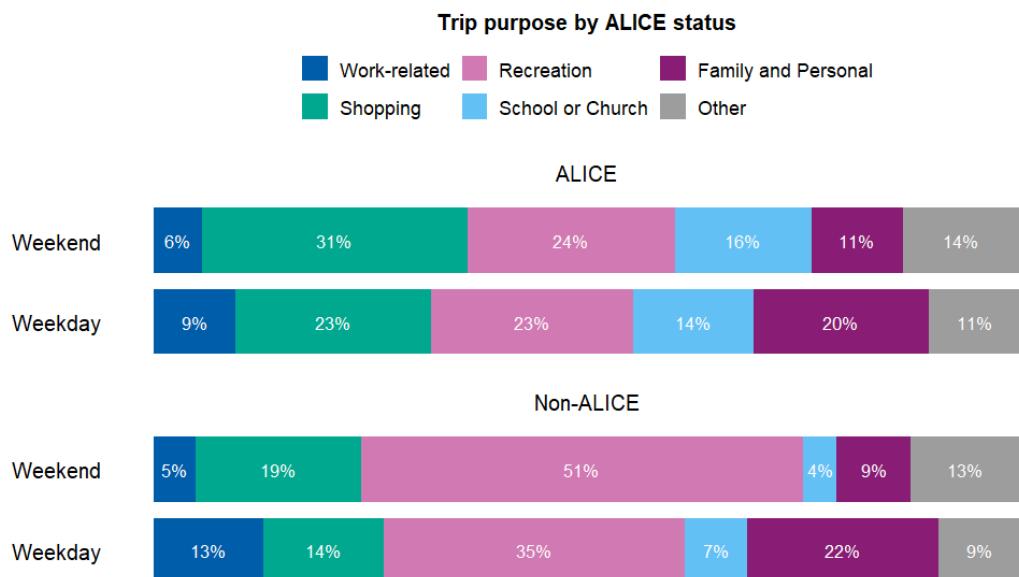


Figure 4-7. Share of Nonmotorized Trip Purpose by Day of Week and ALICE Status in NYS

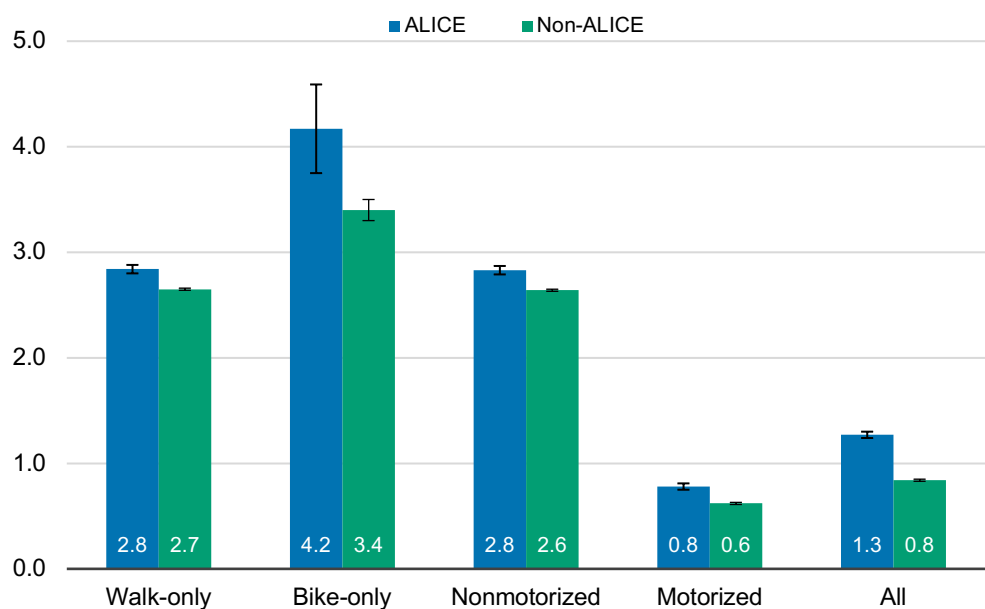


Figure 4-8. Nonmotorized Trip Rate by ALICE Status and User Group in NYS

4.5.1 Household Lifecycle

To further explore how household lifecycle influences nonmotorized travel behaviors of ALICE households, Figure 4-9 breaks down nonmotorized trip rates across four lifecycle categories—single adults without children, multiple adults without children, households with children, and retired households. Among single adults without children, both ALICE and non-ALICE individuals show high

levels of nonmotorized travel, with trip rates nearly identical. This suggests that in the absence of dependents, nonmotorized travel is a preferred or necessary mode for individuals regardless of income.

“Multiple adults, no children” category shows significant difference in bike-only trips between the ALICE groups. ALICE households report a substantially higher bike-only trip rate of 5.9 compared to just 3.7 among non-ALICE. Despite this gap in biking, the overall nonmotorized trip rates remain similar between the two groups. In households with children, overall nonmotorized trip rates drop for both ALICE and non-ALICE groups compared to other lifecycles. ALICE individuals in this group report 2.6 nonmotorized trips per person, slightly higher than 2.5 among non-ALICE. Retired households also demonstrate a notable contrast between ALICE and non-ALICE individuals. ALICE retirees record 3.0 nonmotorized trips per person, compared to 2.7 among non-ALICE. Bike-only trip rates are also higher for ALICE retirees (3.2 vs. 2.3), reinforcing the broader trend that ALICE populations are more active in nonmotorized travel across lifecycle stages.

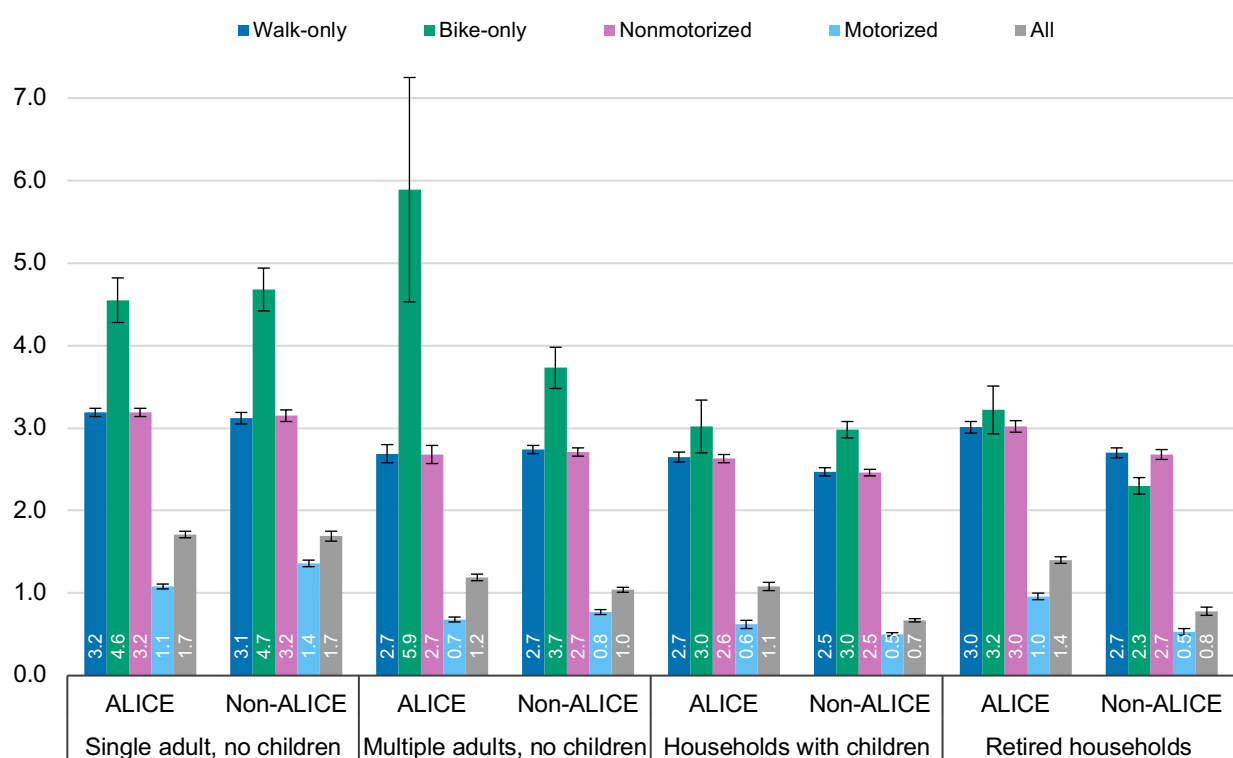


Figure 4-9. Nonmotorized Trip Rate by Household Lifecycle, ALICE Status and User Group in NYS

4.5.2 Household Race

Figure 4-10 presents the variation in nonmotorized trip rates across different racial groups, stratified by ALICE and non-ALICE household status. A consistent trend emerges across all groups: individuals in ALICE households tend to have higher nonmotorized trip rates than their non-ALICE counterparts. Among White households, the difference is most pronounced in the bike-only group. ALICE White individuals report a significantly higher bike-only trip rate (5.0 trips per person), compared to 3.3 for non-ALICE Whites. This bike-oriented travel behavior also drives the total nonmotorized trip rate among

ALICE White households to be the highest of all subgroups (3.0 trips), indicating a possible reliance on bicycles for daily travel needs.

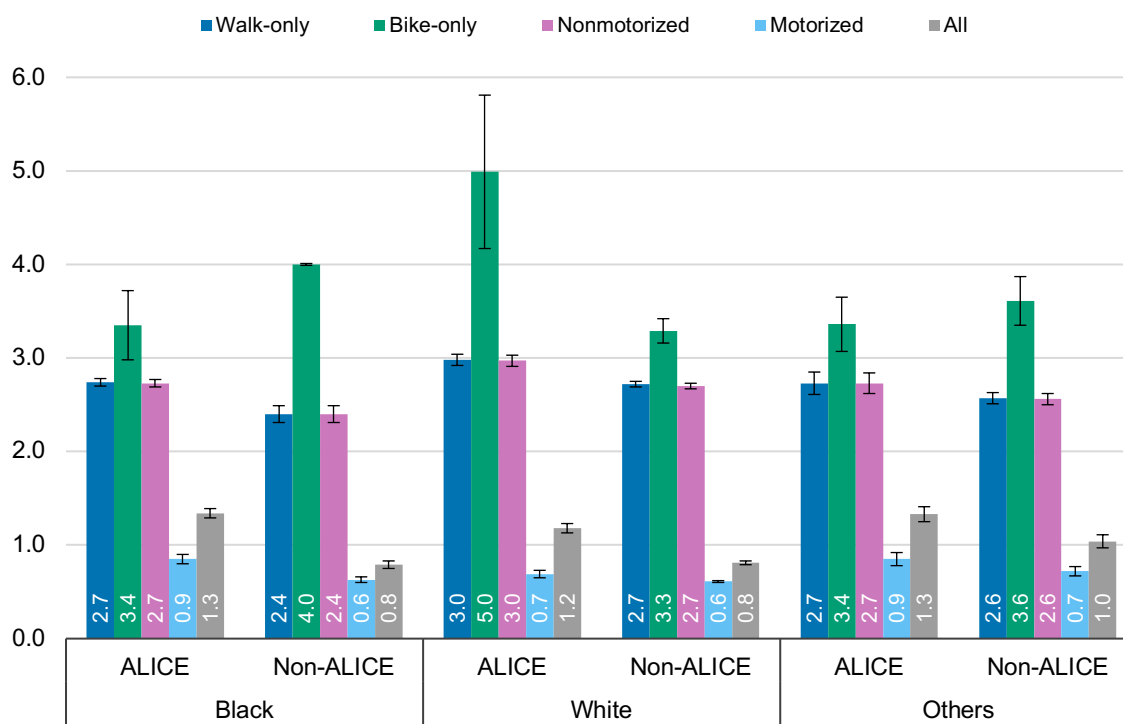


Figure 4-10. Nonmotorized Trip Rate by Household Race, ALICE Status and User Group in NYS

4.5.3 Vehicle Ownership

Vehicle ownership plays a critical role in shaping nonmotorized travel behavior, and Table 4.1 reveals distinct patterns between ALICE and non-ALICE households across varying levels of household vehicle ownership. As expected, individuals in zero vehicle households reported the highest nonmotorized trips, highlighting a clear substitution effect where walking and biking are used in place of unavailable motorized modes. This pattern is observed in both ALICE and non-ALICE households, but the effect is particularly prominent among ALICE zero vehicle households, who report an average of 3.1 nonmotorized trips per person, slightly higher than the 3.0 reported by non-ALICE zero vehicle individuals. Furthermore, ALICE zero vehicle individuals also show a notably high bike-only trip rate of 4.7, the highest across all subgroups.

As vehicle ownership increases, nonmotorized trip rates decline across the board, with the largest drop observed in the two vehicles and more than two vehicle households. For instance, among ALICE households with two vehicles, the nonmotorized trip rate drops to just 2.0, while for those with more than two vehicles, it stabilizes at 2.5. In one-vehicle households, non-ALICE individuals report slightly higher nonmotorized trip rates (2.7) than their ALICE counterparts (2.4).

Table 4-1. Nonmotorized Trip Rate by Household Vehicle Ownership, ALICE Status and User Group in NYS

Household Vehicle Ownership	ALICE Status	Estimates					MOE				
		Walk-only	Bike-only	NMT	MT	All	Walk-only	Bike-only	NMT	MT	All
Zero vehicle	ALICE	3.1	4.7	3.1	1.5	2.1	0.06	0.5	0.05	0.06	0.06
	Non-ALICE	3.0	3.6	3.1	1.7	2.2	0.04	0.27	0.03	0.04	0.04
One vehicle	ALICE	2.5	3.2	2.4	0.6	0.9	0.07	0.37	0.06	0.03	0.04
	Non-ALICE	2.7	3.9	2.7	0.9	1.3	0.05	0.11	0.04	0.03	0.04
Two vehicles	ALICE	2.0	1.8	2.0	0.3	0.4	0.04	0.16	0.04	0.02	0.03
	Non-ALICE	2.3	2.6	2.3	0.4	0.5	0.05	0.12	0.04	0.02	0.02
More than two vehicles	ALICE	2.5	2.8	2.5	0.2	0.4	0.18	0.26	0.16	0.02	0.04
	Non-ALICE	2.2	2.4	2.2	0.3	0.3	0.10	0.07	0.10	0.02	0.02

NMT = Nonmotorized, MT = Motorized, MOE = Margin of Error

4.5.4 Home Ownership

Figure 4-11 shows nonmotorized trip rates by homeownership status across ALICE and non-ALICE households, disaggregated by user group. This comparison provides insight into how economic status and housing tenure intersect to shape travel patterns. Among homeowners, ALICE and non-ALICE households display broadly similar levels of walking trips, averaging around 2.5 to 2.6 trips per person per day. However, the contrast is sharper in the bike-only group: ALICE homeowners report a higher average (3.6 trips) compared to 3.0 for non-ALICE. This difference is statistically significant. When looking at overall nonmotorized rates, the gap is narrower but still leans slightly higher for ALICE households. For motorized-only users, both ALICE and non-ALICE homeowners report low nonmotorized trip rates, suggesting they incorporate some walking or biking alongside car use. Renters show a distinct pattern. Both ALICE and non-ALICE renters take more nonmotorized trips than homeowners, underscoring how renting often coincides with denser, more urbanized settings that support walking and biking. Walk-only and nonmotorized trip rates are higher for renters across both economic groups compared to owners. For motorized-only users, however, non-ALICE renters record slightly higher nonmotorized activity compared to ALICE renters.

4.5.5 Driver Availability

Figure 4-12 compares nonmotorized trip rates across ALICE and non-ALICE households based on the number of licensed drivers available in the household. Among households with only one driver, trip rates are relatively similar across ALICE and Non-ALICE groups. Walk-only and nonmotorized users in both groups record comparable rates, suggesting that limited access to drivers encourages higher reliance on walking and biking regardless of ALICE status. However, bike-only users in non-ALICE households make slightly more trips than ALICE households. The differences become more pronounced in households with two drivers. For households with more than two drivers, both groups show a decline in nonmotorized travel, but the drop is steeper for non-ALICE households. ALICE households still maintain

higher walk-only and bike-only rates, whereas non-ALICE households record some of the lowest nonmotorized trip rates across all categories.

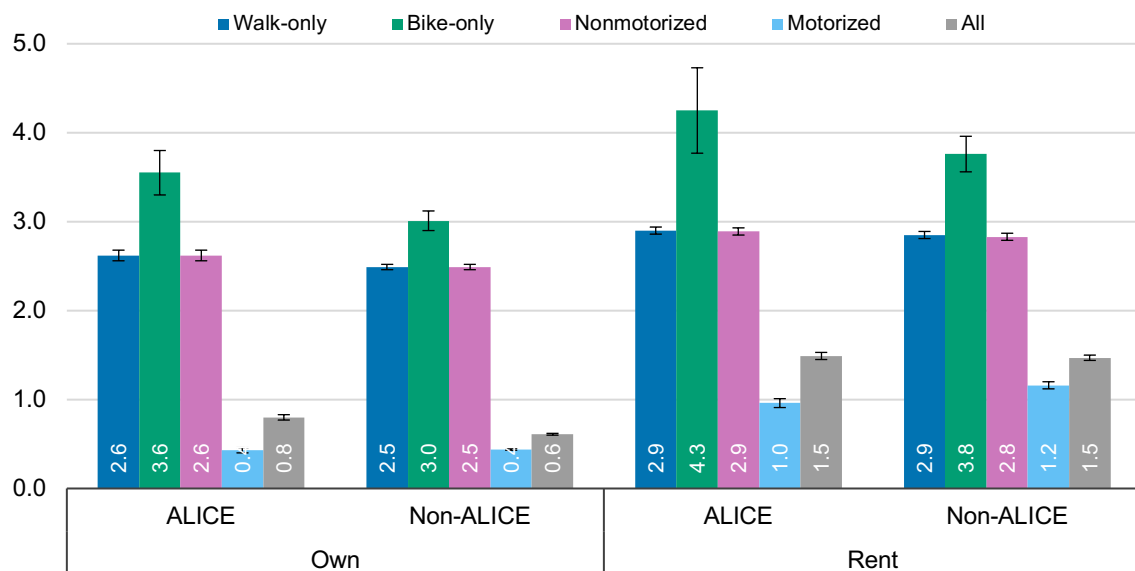


Figure 4-11. Nonmotorized Trip Rate by Home Ownership, ALICE Status and User Group in NYS

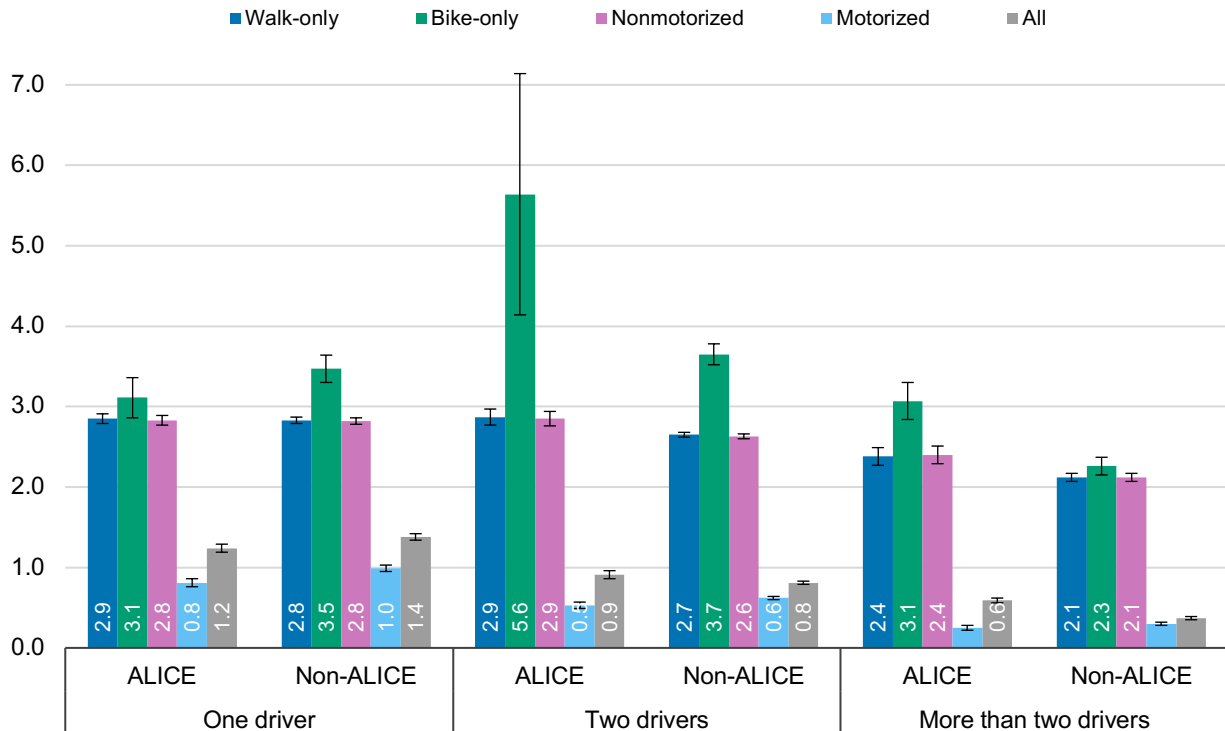


Figure 4-12. Nonmotorized Trip Rate by Driver Availability, ALICE Status and User Group in NYS

Overall, the results emphasize that ALICE households consistently make more nonmotorized trips than non-ALICE households across all levels of driver availability. This indicates that even when vehicles and drivers are available, ALICE households may continue to rely more on walking and biking, possibly reflecting differences in income, car ownership quality, or access to alternative transportation.

4.5.6 Carshare

Figure 4-13 highlights how carshare participation influences nonmotorized trip-making across ALICE and non-ALICE households. For ALICE households that use carshare, nonmotorized travel rates are high. This indicates that carshare households in the ALICE group rely heavily on active modes of transport. Conversely, non-ALICE households with carshare access show more modest trip-making. Their walk-only and bike-only trip rates are lower, with bike-only use averaging around 3 trips, much closer to non-ALICE households without carshare. When looking at households without carshare, both ALICE and Non-ALICE households record substantially lower nonmotorized rates compared to their carshare-using counterparts. Still, ALICE households continue to make slightly more walk-only and overall nonmotorized trips than non-ALICE households.

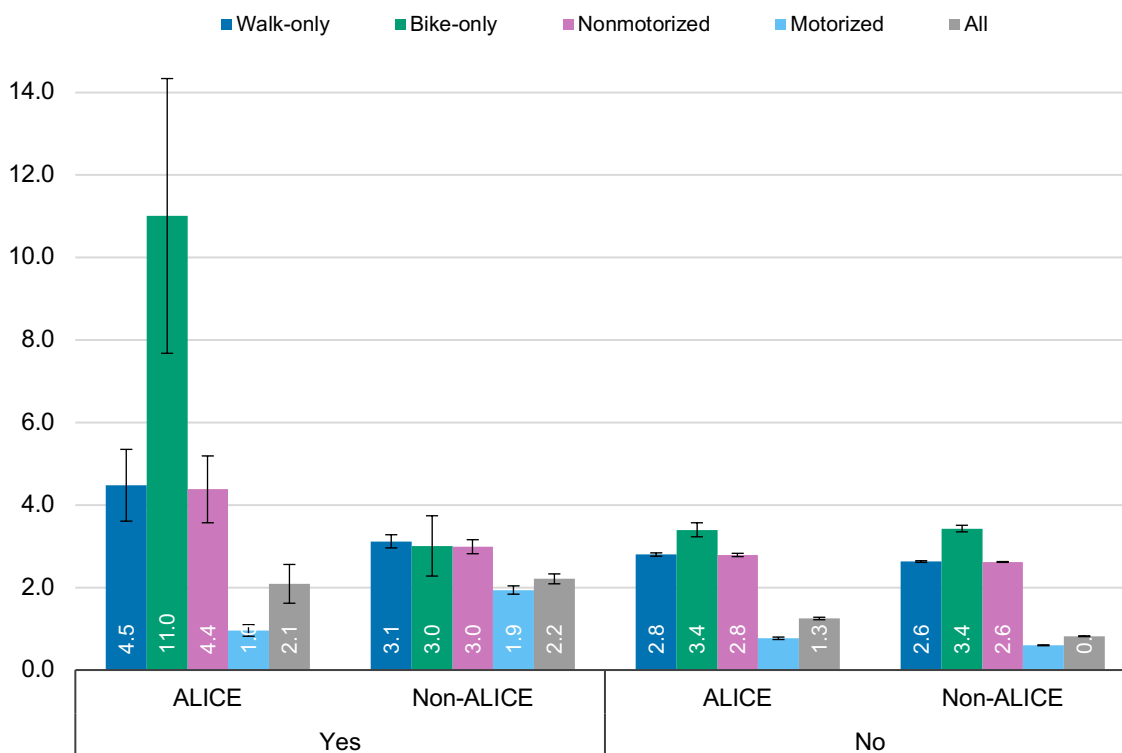


Figure 4-13. Nonmotorized Trip Rate by Carshare, ALICE Status and User Group in NYS

4.5.7 Rideshare

Figure 4-14 compares nonmotorized trip rates between ALICE and non-ALICE households based on whether they participated in ridesharing, across the different user groups. For walk-only users, both ALICE and non-ALICE households show higher trip rates when they participate in ridesharing compared

to when they do not. ALICE households average just over 3.1 daily walk trips under rideshare participation versus under 2.8 otherwise. These differences are statistically significant. This suggests that rideshare participation is associated with more walking, potentially because rideshare users rely on walking to connect to pick-up and drop-off points.

In the nonmotorized category overall, both ALICE and non-ALICE households record higher trip rates with rideshare use. The motorized group (those who make motorized trips but are also observed taking some nonmotorized trips) reveals a larger gap. ALICE households with rideshare average 1.3 nonmotorized trips, compared to 0.7 without rideshare. Non-ALICE households also show a notable increase, from 0.5 to 1.4 trips. This suggests that rideshare participation may facilitate more mixed-mode travel, where individuals combine motorized services with walking or biking.

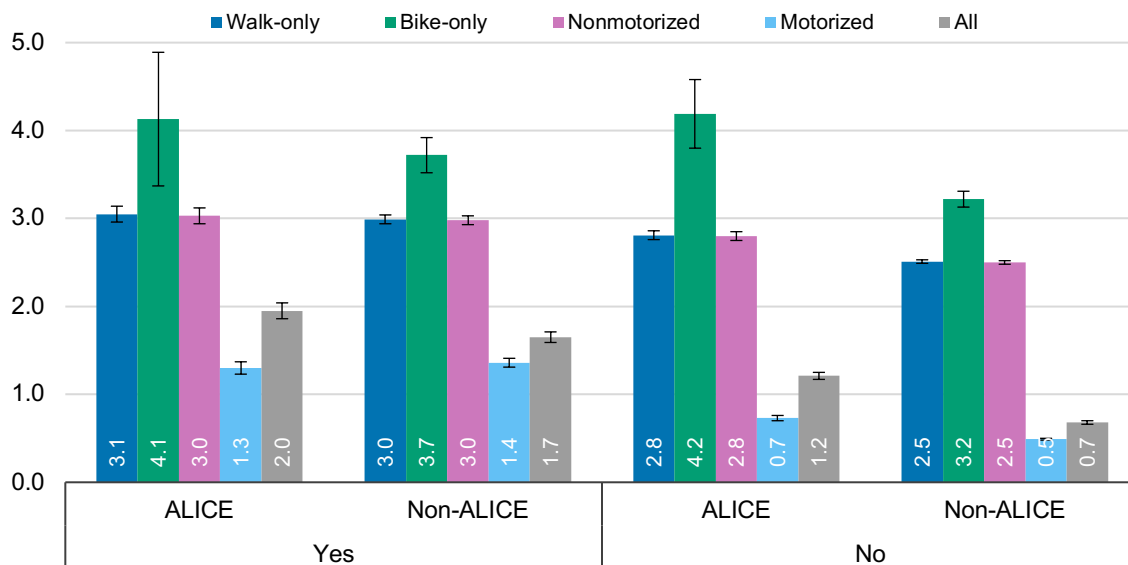


Figure 4-14. Nonmotorized Trip Rate by Rideshare, ALICE Status and User Group in NYS

5. CONCLUSIONS

Nonmotorized travel (walking and biking) constitutes a substantial portion of all US travel and is recognized as an integral component of comprehensive mobility systems. While active transportation modes promote accessibility, affordability, and public health outcomes, these benefits are found to vary significantly across geographic, demographic, and socioeconomic contexts. To examine these dynamics within NYS, this analysis utilized 2017 NHTS add-on data. The primary research objectives were to address the systematic "4Ws" framework of nonmotorized travel—*Who* participates, *Where* trips occur, *When* they are undertaken, and *Why* they are made—while conducting user segmentation analysis and comparative examination of travel behavior between ALICE and non-ALICE household categories.

The findings reveal that nonmotorized trips are found to constitute one-quarter of all travel in NYS, a share substantially exceeding the national average. These trips are observed to concentrate strongly in NYC, which accounts for more than 70% of the state's walking and biking activity. Demographically, middle-aged adults (40–64 years) are found to represent the largest participant group, particularly in biking, where they comprise nearly half of all users. Seniors and children, by contrast, are observed to report substantially fewer trips. Health and physical activity status also play critical roles—those with higher self-reported health and more frequent exercise are found to exhibit higher rates of walking and biking. Temporal patterns highlight that nonmotorized travel differs from commuting modes, with midday peaks on weekdays and extended afternoon activity on weekends, often tied to discretionary or recreational purposes. Recreational activities are found to represent the most common purpose for nonmotorized trips across all regions and home geographies.

Segmentation analysis of user groups reveals different participation patterns by sex: women are found to dominate walk-only travel, while biking is observed to be disproportionately male-dominated. Households with children consistently represent the largest share across all groups, suggesting that family responsibilities shape active travel behavior. Zero vehicle households are found to report significantly higher walking frequencies across both NYC and remaining NYS regions, underscoring the relationship between vehicle availability and reliance on nonmotorized modes. The comparison of ALICE and non-ALICE households highlights notable differences in travel reliance: ALICE households are observed to consistently make more nonmotorized trips, with particularly substantial gaps outside NYC, where non-ALICE households are found to report far fewer walking or biking trips. These differences extend into trip purposes, with ALICE households observed to show higher concentrations of morning school- and church-related trips, while non-ALICE households are found to exhibit midday peaks associated with family and personal activities.

The implications of these findings extend beyond descriptive statistics into transportation planning and modeling applications. By documenting who walks and bikes, when these trips occur, and for what purposes, this study provides analytical inputs that can inform four-step travel demand models, activity-based models, and discrete choice modeling frameworks. Active transportation has historically been underrepresented in forecasting methodologies, often treated as residual trips rather than integral travel behavior. The evidence presented here demonstrates the necessity of incorporating nonmotorized trips as fundamental components of regional travel demand, both in terms of person-trip generation and spatiotemporal distribution patterns. Enhanced representation of walking and biking in modeling frameworks can improve system-level assessments of accessibility, mode choice calibration, and overall network performance across both dense urban and dispersed non-urban contexts.

Several limitations should be acknowledged. First, the analysis relies on a single cross-sectional dataset (2017 NHTS), which does not capture temporal shifts in behavior that may have occurred in subsequent years, including changes accelerated by the COVID-19 pandemic. Second, while the survey provides robust statewide representation, smaller subgroups (such as bike-only users) have relatively limited

sample sizes, constraining the statistical power of detailed disaggregated analysis. Third, the survey depends on self-reported information, which may be subject to recall bias or underreporting of short, incidental walk trips. Future research could address these limitations by combining survey data with passive mobility data sources such as smartphone GPS or fitness applications and applying longitudinal methods to track behavioral shifts over time. Additionally, incorporating built environment measures and spatial accessibility indicators would enable robust linkages between land use, infrastructure, and nonmotorized travel patterns.

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APPENDIX A. GLOSSARY OF NHTS TERMS

APPENDIX A. GLOSSARY OF NHTS TERMS

This glossary provides the most commonly used terms in the National Household Travel Survey (NHTS) and the terms used in this report, along with definitions. These definitions are provided to assist the user in interpreting the NHTS data.

Adult	For NHTS, this is defined as a person 18 years or older.
Block Group	A subdivision of a census tract that contains 600–3,000 people. The source used for the 2017 NHTS was the United States Census Bureau 2014 TIGER/Line Shapefiles (derived from Census 2010 definition).
Census Tract	A small subdivision of a county, generally having a population size between 1,200 and 8,000 people, with an optimum size of 4,000 people. The geographic size of the tract may vary considerably, depending on population density. Tracts were designed to be homogeneous regarding population characteristics, economic status, and living conditions when they were first delineated. Because the first tracts were delineated for the 1890 Census, current tracts may be far from homogeneous. The source used for the 2017 NHTS was the United States Census Bureau 2014 TIGER/Line Shapefiles (derived from Census 2010 definition).
Destination	For travel day trips, the destination is the point at which there is a break in travel, except if the break is only to change vehicles or means of transport.
Elderly	A person age 65 years or older.
Elderly Household	Households with one or more persons age 65 years or older.
Employed	A person is considered employed if that person worked for pay, either full time or part time, during the week before the interview. This includes persons who work at home and/or have more than one job.
Household	A group of persons whose usual place of residence is a specific housing unit; these persons may or may not be related to each other. The total of all US households represents the total civilian noninstitutionalized population. A household does not include group quarters (i.e., 10 or more persons living together, none of whom are related).
Household Income	Household income is the money earned by all family members in a household, including those temporarily absent. Annual income consisted of the income earned 12 months preceding the interview. Household income includes monies from all sources, such as wages and salary, commissions, tips, cash bonuses, income from a business or farm, pensions, dividends, interest, unemployment or worker's compensation, social security, veterans' payments, rent received from owned property (minus the operating costs), public assistance payments, regular gifts of money from friends or relatives not living in the household, alimony, child support, and other kinds of periodic money income other than earnings. Household income excludes in-kind income such as room and board, insurance payments, lump-sum inheritances, occasional gifts of money from persons not living in the same household, withdrawal of savings from banks, tax refunds, and the proceeds of the sale of one's house, car, or other personal property.

Household Members	Household members include all people, whether present or temporarily absent, whose usual place of residence is in the sample unit. Household members also include people staying in the sample unit who have no other usual place of residence elsewhere.
Household Vehicle	A household vehicle is a motorized vehicle that is owned, leased, rented, or company-owned and available to be used regularly by household members during the 2 week travel period. Household vehicles include vehicles used solely for business purposes or business-owned vehicles if they are driven home and can be used for the home-to-work trip (e.g., taxicabs and police cars). Household vehicles include all vehicles that were owned or available for use by members of the household during the travel period, even though a vehicle may have been sold before the interview. Vehicles excluded from household vehicles are (1) those that were not working and were not expected to be working within 60 days and (2) those that were purchased or received after the designated travel day.
Journey-to-Work Trips (Commute Trips)	Includes travel to and from a place where one reports for work. Does not include any other work-related travel. Does not include trips for persons who work at home.
Means of Transportation	A mode of travel used for going from one place (origin) to another (destination). A means of transportation includes private and public transit modes, as well as walking. The following transportation modes, grouped by major mode, are included in the NHTS data. Active modes—include modes by which a person must actively move from one place to the next. • Walk: This category includes walking and jogging. • Bicycle: This category includes bicycles of all speeds and sizes that do not have a motor. Private vehicle—a stipulation for being a private vehicle is that the vehicle is privately operated, including rental cars. • Car: Includes cars and station wagons. Leased and rented cars are included if privately operated and not used for picking up passengers in return for fare. • Sport utility vehicle: Includes vehicles that are a hybrid of design elements from a van, pickup truck, and station wagon. Examples include a Ford Explorer, Jeep Cherokee, or Nissan Pathfinder. • Van: Includes vans or minivans designed to carry 5–13 passengers or to haul cargo. • Pickup truck: Includes vehicles with an enclosed cab that usually accommodates two to three passengers and has an open cargo area in the rear. Late-model pickups often have a back seat that allows for total seating of four to six passengers. Pickup trucks usually have a wheelbase the same size as that of a full-size station wagon. This category also includes pickups with campers. • Motorcycle/moped: This category includes large, medium, and small motorcycles and mopeds.

- RV (motor home, ATV, snowmobile): An RV or motor home includes a self-powered recreational vehicle that is operated as a unit without being towed by another vehicle (e.g., a Winnebago motor home). This category also includes ATVs and snowmobiles.

Rental car—Includes Zipcar and Car2Go, in addition to commercially rented cars for private use.

- Golf cart/Segway: This includes all electric or gas-operated vehicles designed for use on a golf course, but whose use has recently extended to within smaller, often gated, communities. NOTE: Travel taken via golf cart is NOT included in vehicle trips and vehicle miles of travel.

Public transportation—as used in Federal Highway Administration publications and analysis of NHTS data, typically includes the following (indicated in italics in the following text): public or commuter bus, commuter rail, and subway/elevated rail/light rail/streetcar.

Bus—This category includes:

- *Public or commuter buses*: local public transit buses available to the public and buses used for short-distance public transport purposes (e.g., city bus or public bus), school buses
- Private/charter/tour/shuttle buses: private buses operating on a fixed schedule between population centers and buses that shuttle passengers from one fixed place to another (e.g., airport shuttles)
- City-to-city buses: buses that run from one urban center to another (e.g., Greyhound)

Train—This category includes:

- *Amtrak/commuter rail*: trains that run from one urban center to another
- *Subway/elevated rail/light rail/streetcar* (also known as rail rapid transit): a high-capacity system operated on a fixed-rail or guideway system on a private right of way, and vehicles that run on a fixed-rail system powered by electricity obtained from an overhead power distribution system

Other modes—This category includes:

- School buses
- Paratransit/dial-a-ride
- Taxi/limo: Use of a taxicab by a passenger for fare, including limousines. In 2017, this category also includes ridesharing such as Uber and Lyft.
- Airplane: includes commercial airplanes and smaller planes available for use by the public in exchange for a fare. Private and corporate planes and helicopters are also included.
- Boat/ferry/water taxi: includes travel by passenger line ferries
- Something else: includes any type of transportation not previously listed (e.g., skateboards, roller blades, sailboats, cruise ships)

Metropolitan Statistical Area (MSA)

Except in the New England states, a metropolitan statistical area is a county or group of contiguous counties that contains at least one city of 50,000 inhabitants or more, or “twin cities” with a combined population of at least 50,000. In addition, contiguous counties are included in an MSA if, according to certain criteria, they are socially and economically integrated with the central city. In the New England states, MSAs consist of towns and cities instead of counties. The

	source used for the 2017 NHTS was the United States Census Bureau 2014 TIGER/Line Shapefiles (derived from the Census 2010 definition).
Motorized Vehicle	Motorized vehicles include all vehicles licensed for highway driving. Snow mobiles and minibikes are specifically excluded.
New York City (NYC)	New York City is defined in this report as the five-county area: Bronx, Kings (Brooklyn), Queens, New York (Manhattan), and Richmond (Staten Island).
Origin	Origin is the starting point of a trip.
Person Trip	A person trip is a trip by one or more persons in any mode of transportation. Each person is considered as making one person trip. For example, four persons traveling together in one auto are counted as four person trips.
Population Density	Category of population density (persons per square mile) in the Census Block Group of the household's home location: 0–99, 100–499, 500–999, 1,000–1,999, 2,000–3,999, 4,000–9,999, 10,000–24,999, and 25,000–999,999.
Privately Owned Vehicle (POV)	A privately owned or privately operated vehicle. In either case, the vehicle is not available to the public for a fee (i.e., it is not a bus, subway, taxi, or other vehicle for which a fee is charged for use).
Travel Day	A travel day is a 24 h period from 4:00 a.m. to 3:59 a.m. designated as the reference period for studying trips and travel by members of a sampled household.
Travel Day Trip	A travel day trip is defined as any time the respondent went from one location to another by private motor vehicle, public transportation, bicycle, walking, or other means during the NHTS-assigned reporting travel day. However, a separate trip is not counted in two instances: 1. When the sole purpose for the trip is to get to another vehicle or mode of transportation to continue to the destination. 2. Travel within a shopping center, mall, or shopping areas of four to five blocks is to be considered as travel to one destination.
Vehicle	In the 2017 NHTS, the term <i>vehicle</i> includes autos, passenger vans, sport utility vehicles, pickups and other light trucks, RVs, motorcycles, and mopeds owned or available to the household.
Vehicle Trip	A trip by a single POV, regardless of the number of persons in the vehicle.
Work-Related Travel	Trips related to business activities except travel to the place of work; for example, a plumber drives to a wholesale dealer to purchase supplies for his business or a company executive travels from his office to another building to attend a business meeting. Business, out-of-town trips, and professional conventions are also included.
Worker	See “Employed.”

